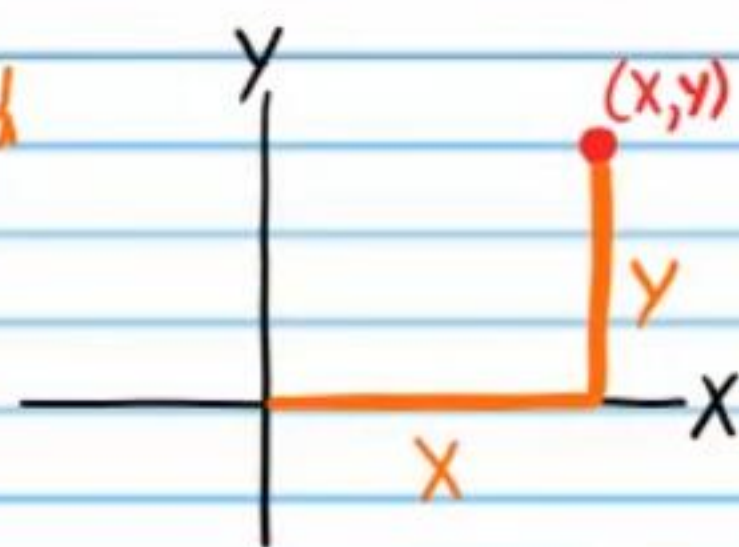


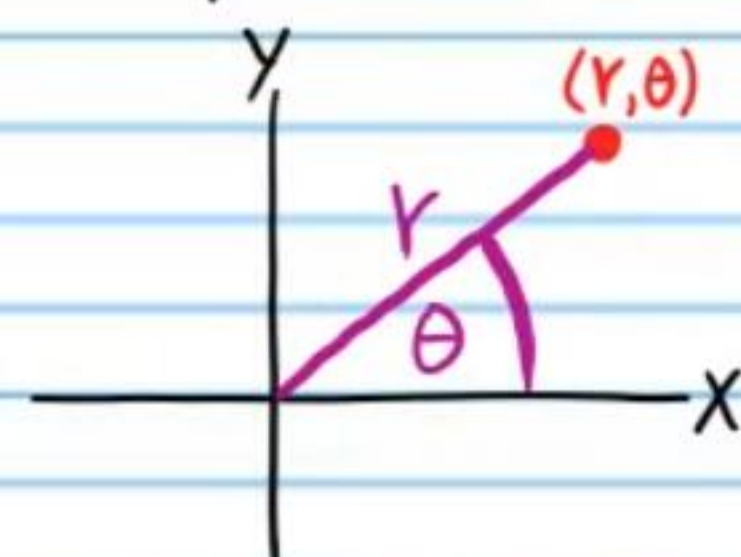
Polar Coordinates

Name
Date
Course

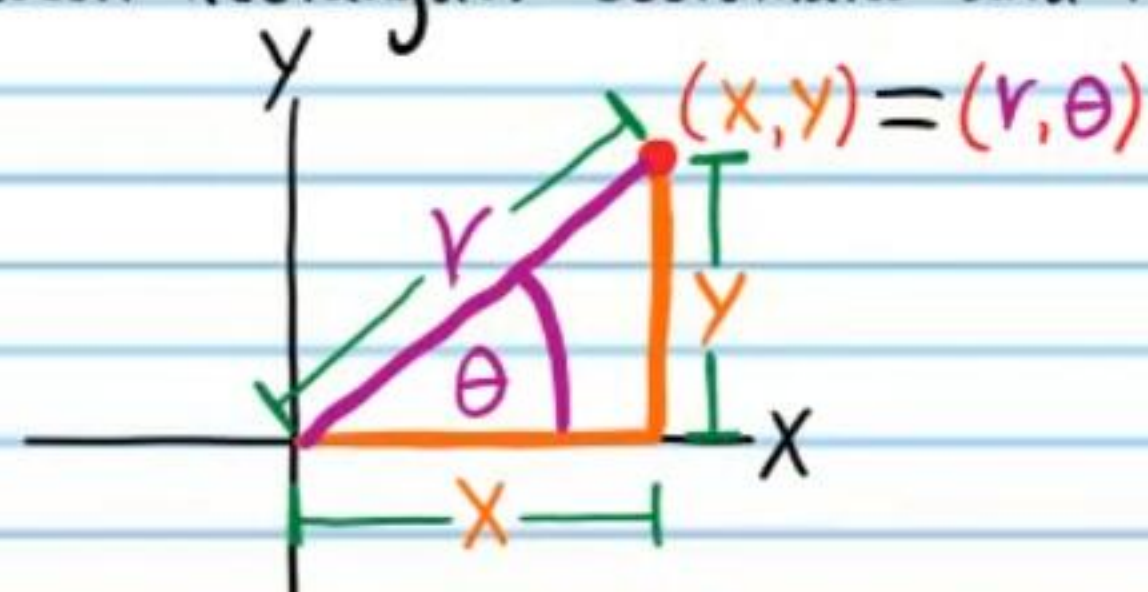
Rectangular Coordinates: Numbers (called x and y values) that are used to plot points x units left or right of the origin and y units above or below the origin.



Polar Coordinates: Numbers (called r and θ values) that are used to plot points r units from the origin at an angle θ units from the positive x -axis.



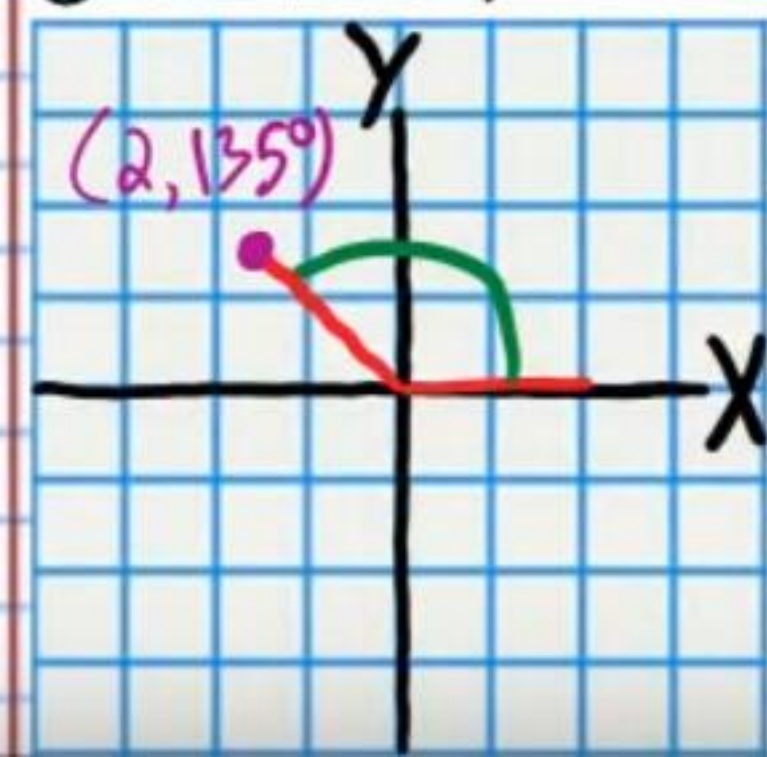
The Relationship between Rectangular Coordinates and Polar Coordinates



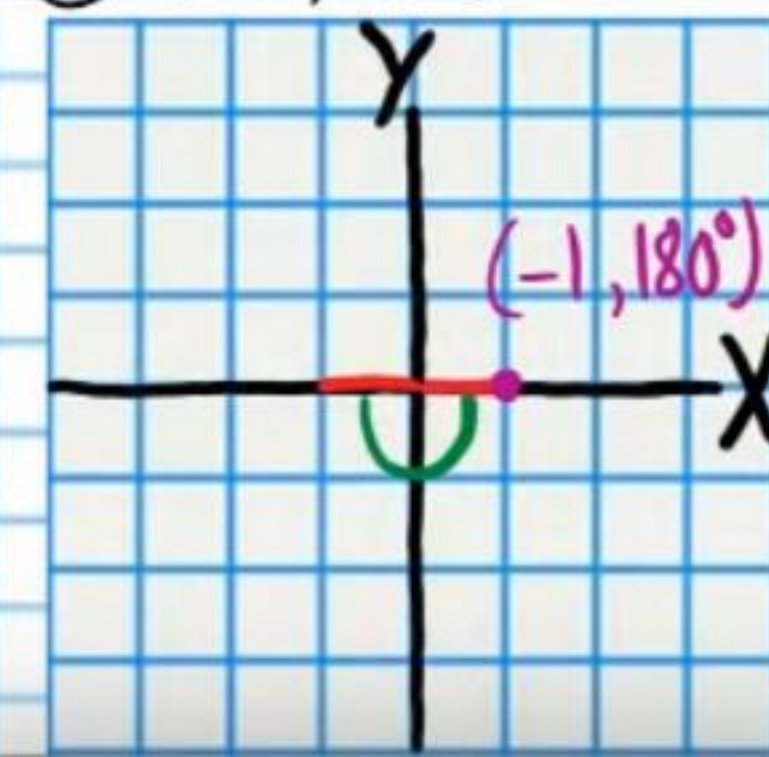
Plotting Points Using Polar Coordinates

Draw the points described by the polar coordinates below.

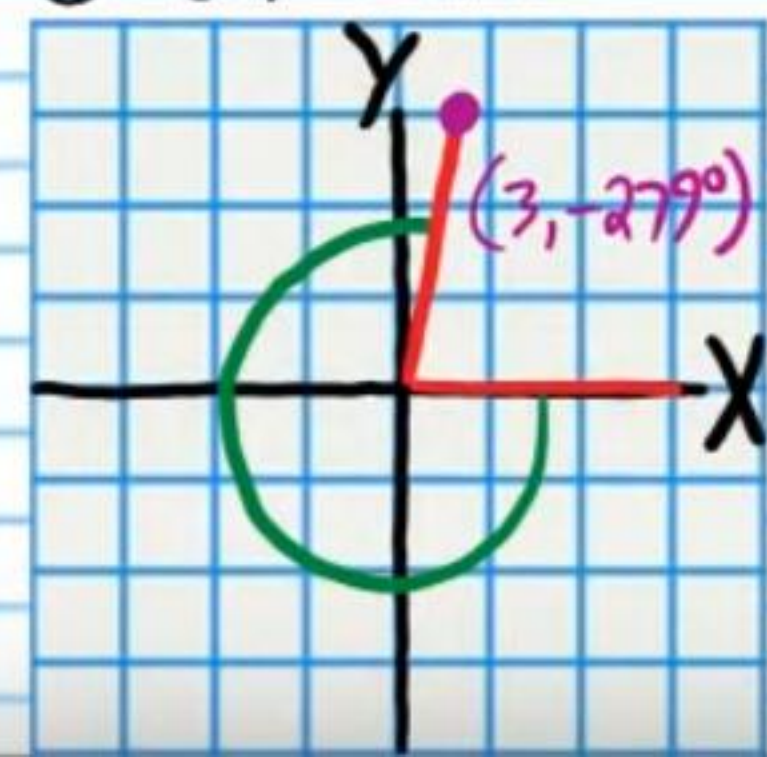
① $(2, 135^\circ)$



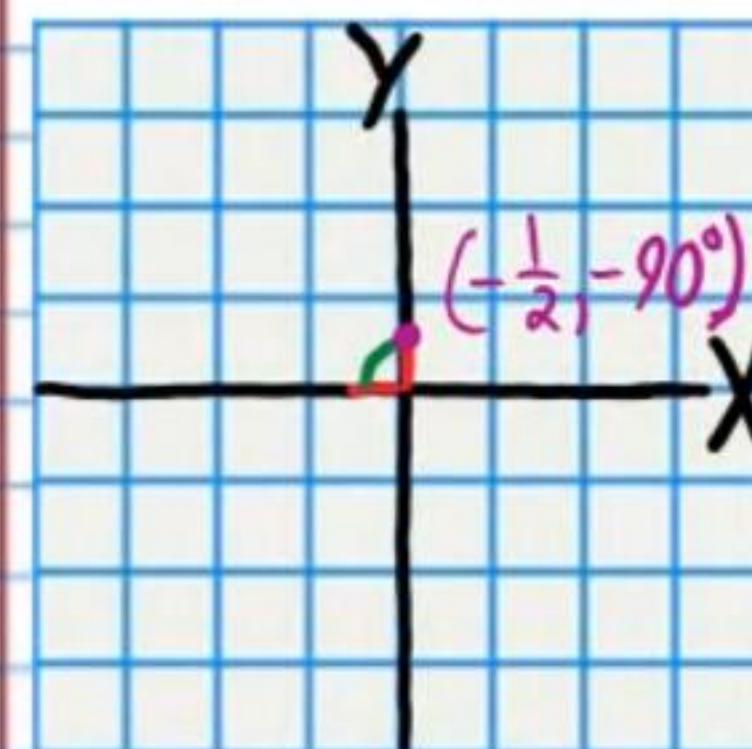
② $(-1, 180^\circ)$



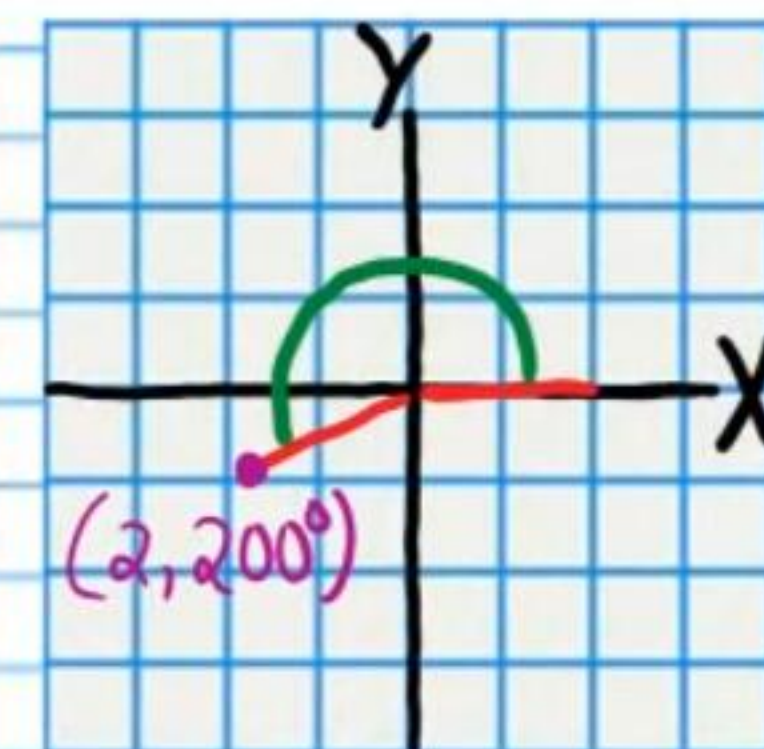
③ $(3, -279^\circ)$



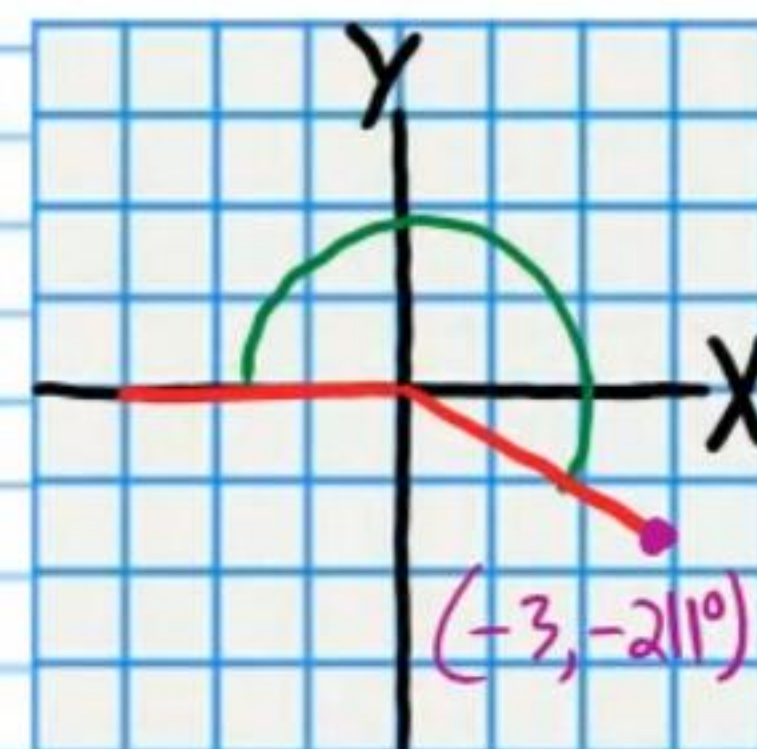
④ $(-\frac{1}{2}, -90^\circ)$



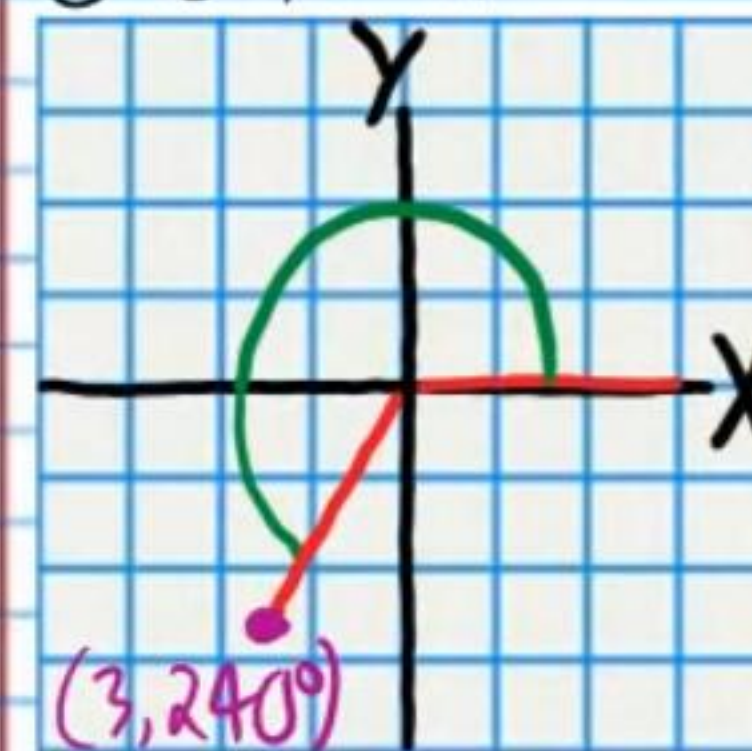
⑤ $(2, 200^\circ)$



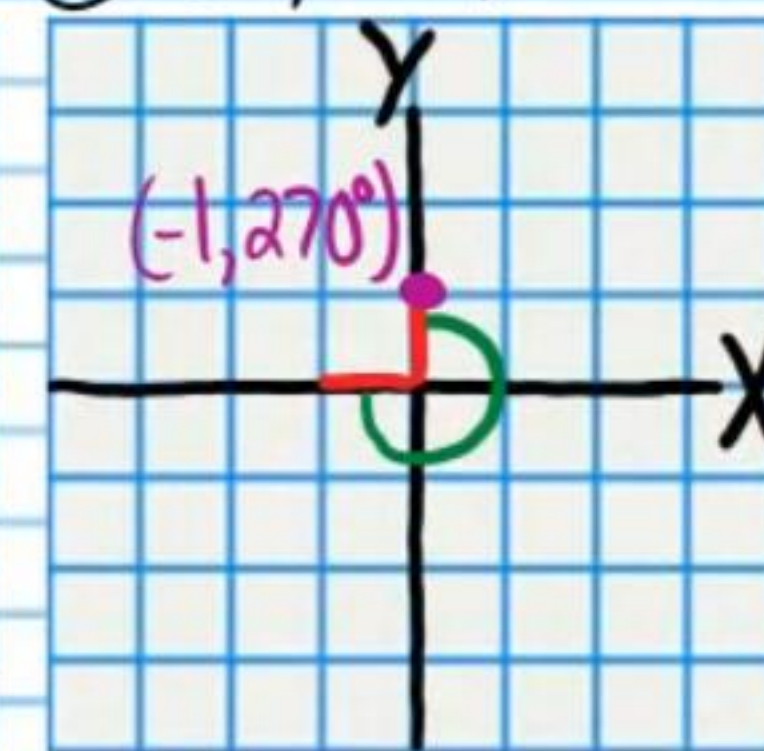
⑥ $(-3, -211^\circ)$



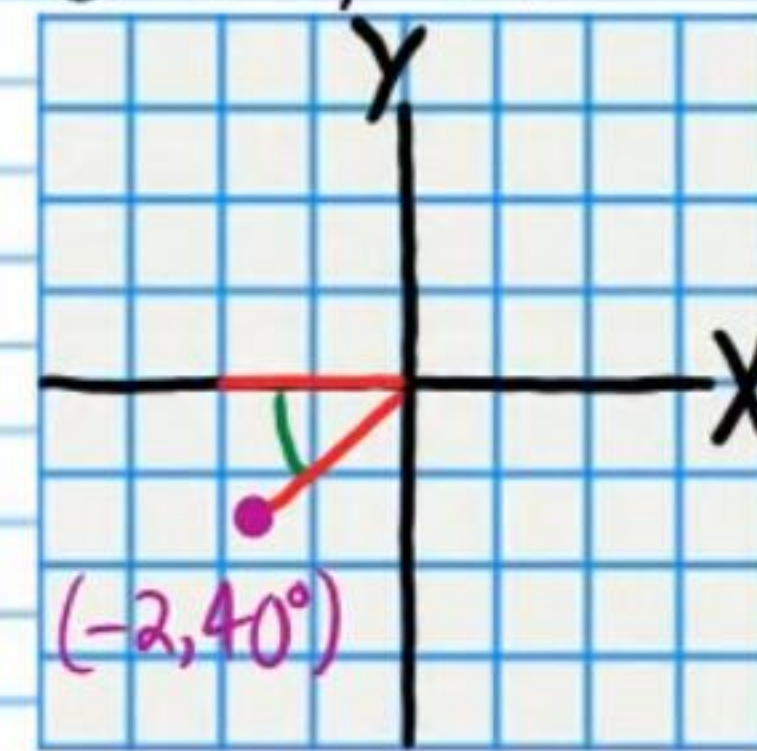
⑦ $(3, 240^\circ)$



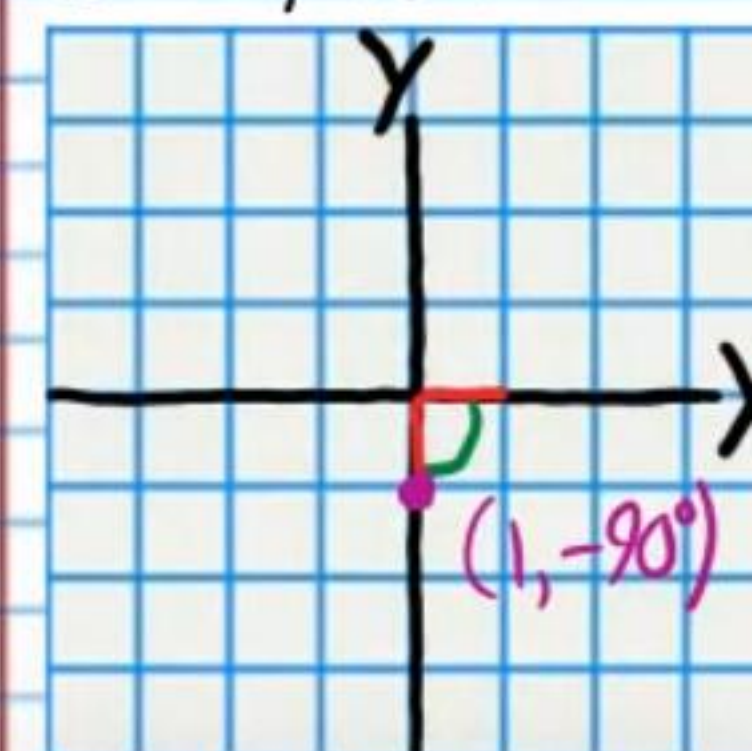
⑧ $(-1, 270^\circ)$



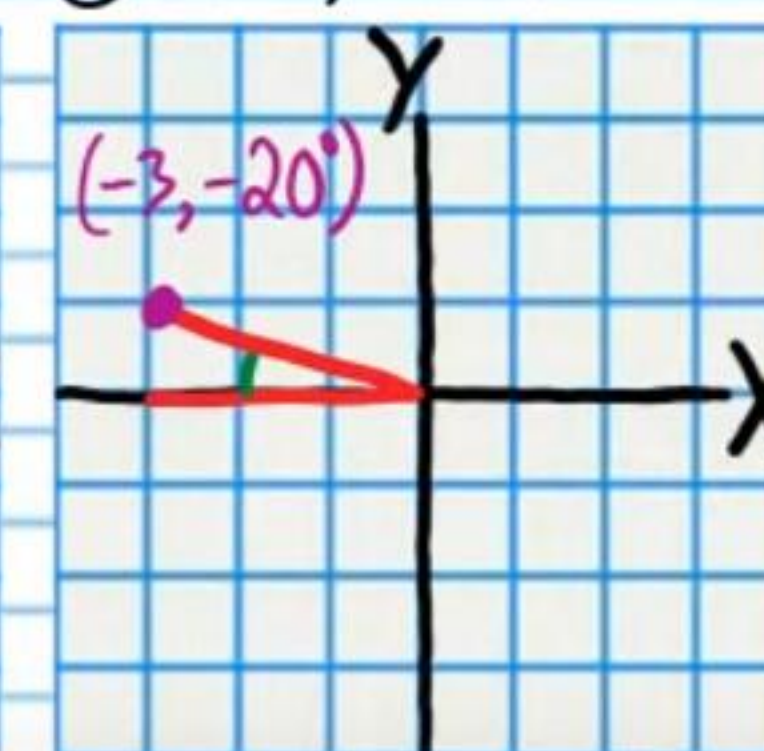
⑨ $(-2, 40^\circ)$



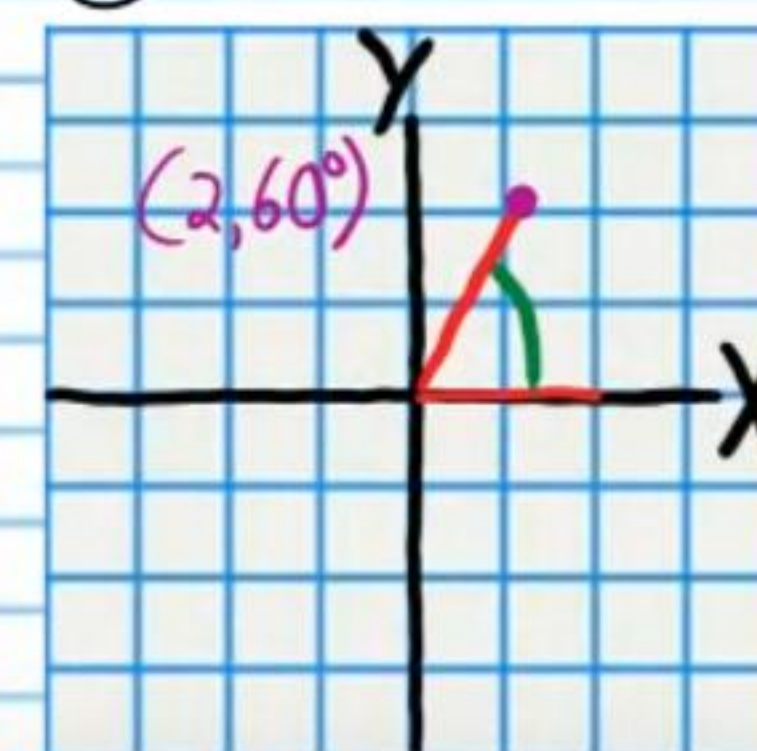
⑩ $(1, -90^\circ)$



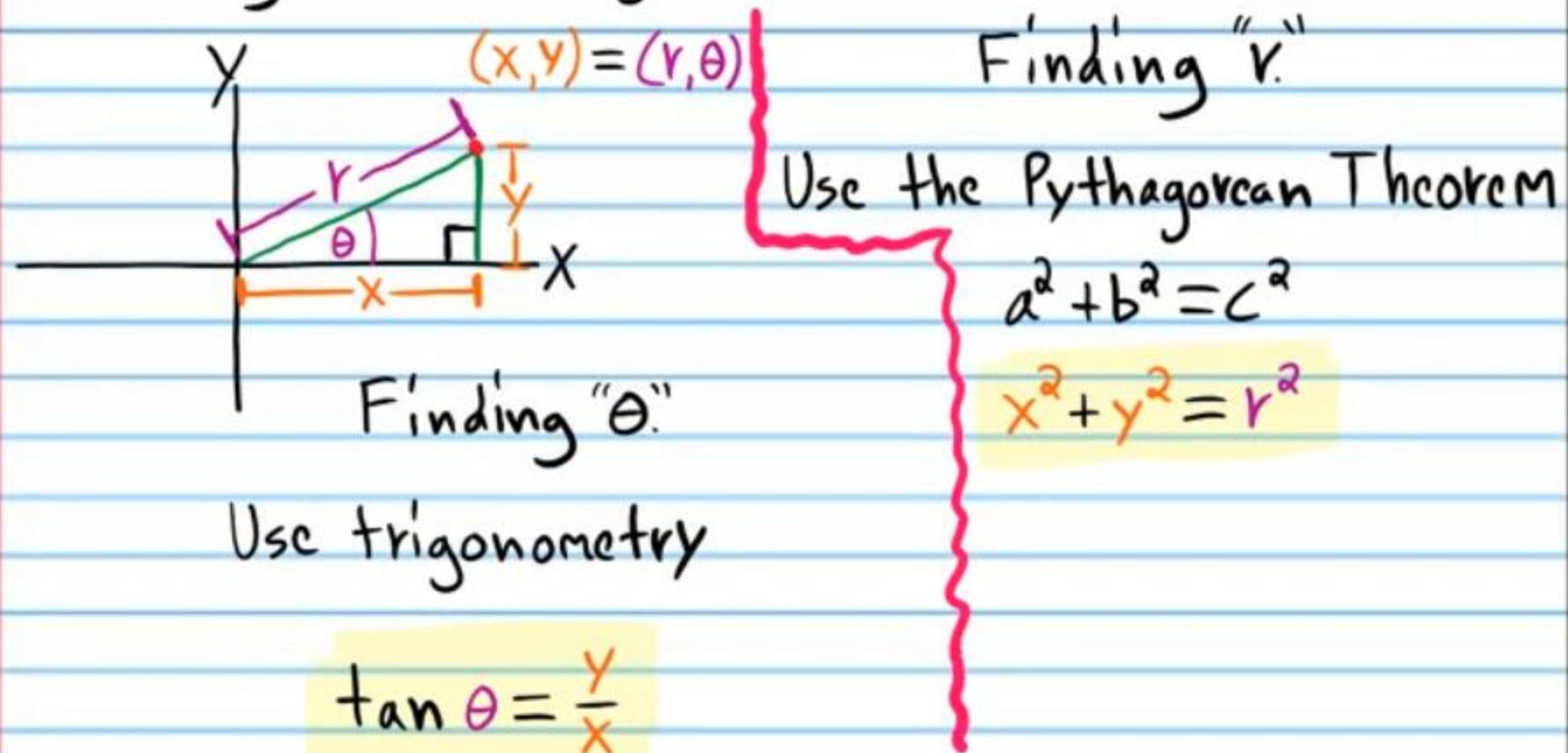
⑪ $(-3, -20^\circ)$



⑫ $(2, 60^\circ)$

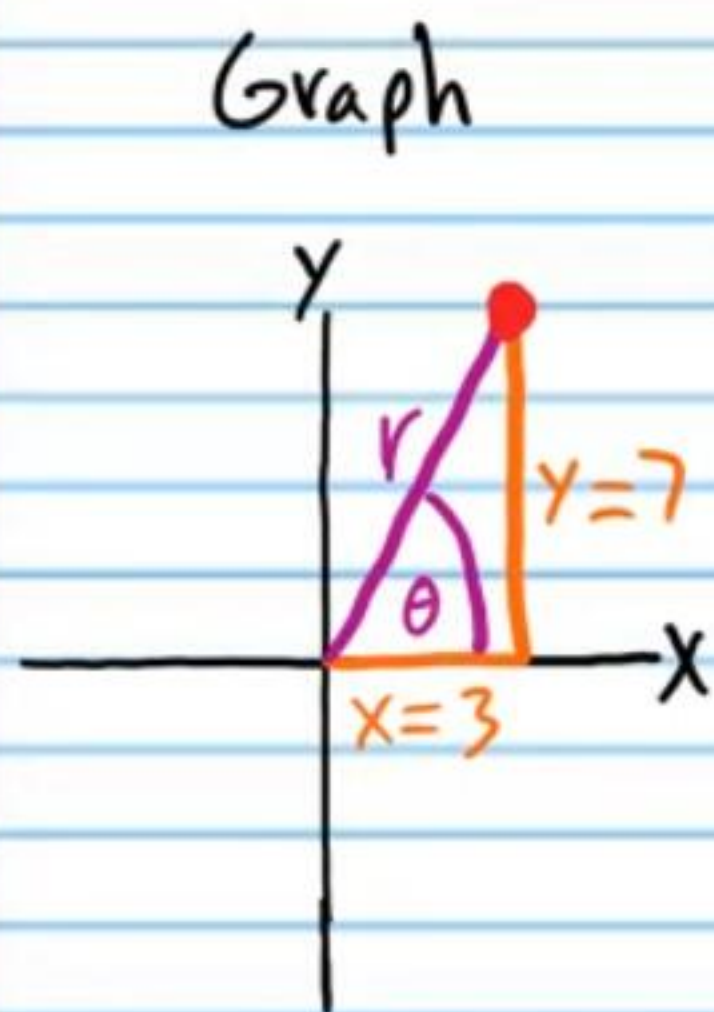


Converting from Rectangular Coordinates to Polar Coordinates



Convert the rectangular coordinates below to polar coordinates.
Use degree units for angle measures. Round to the nearest tenth if necessary.

• (13) (3, 7)



Find r.

$$a^2 + b^2 = c^2$$

$$x^2 + y^2 = r^2$$

$$3^2 + 7^2 = r^2$$

$$9 + 49 = r^2$$

$$r^2 = 58$$

$$r = \pm\sqrt{58}$$

$$(\sqrt{58}, 66.8^\circ)$$

Find theta.

$$\tan \theta = \frac{\text{opp}}{\text{adj}}$$

$$\tan \theta = \frac{y}{x}$$

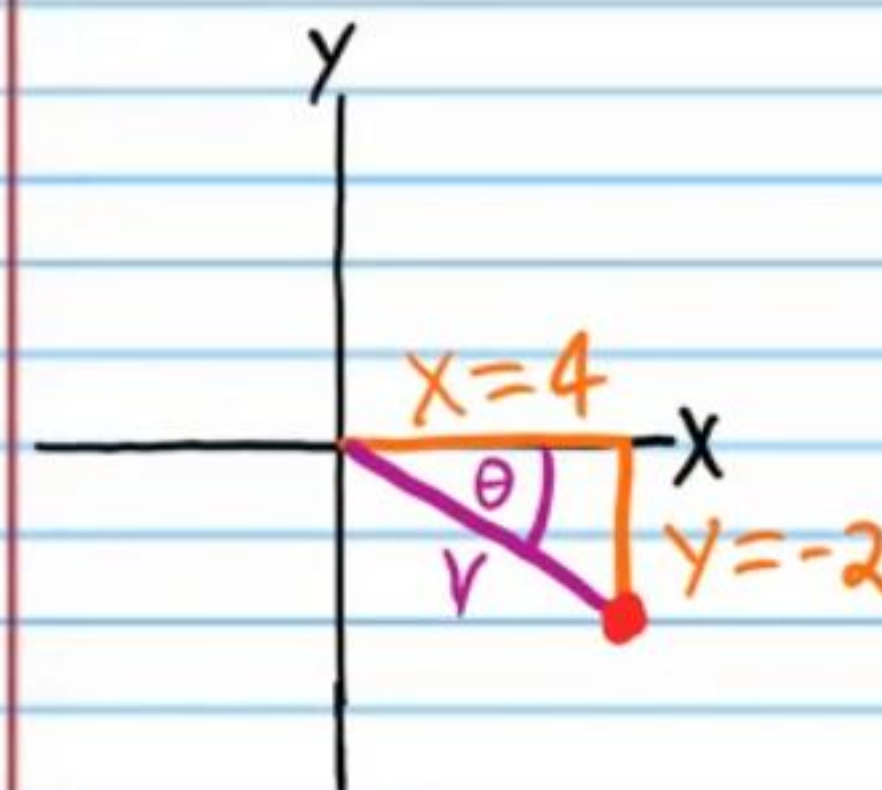
$$\tan \theta = \frac{7}{3}$$

$$\theta = \tan^{-1}\left(\frac{7}{3}\right)$$

$$\theta \approx 66.8^\circ$$

(14) (4, -2)

Graph



Find r.

$$x^2 + y^2 = r^2$$

$$4^2 + (-2)^2 = r^2$$

$$16 + 4 = r^2$$

$$20 = r^2$$

$$r = \pm\sqrt{20}$$

$$r = \pm 2\sqrt{5}$$

$$(2\sqrt{5}, -26.6^\circ)$$

Find theta.

$$\tan \theta = \frac{y}{x}$$

$$\tan \theta = \frac{-2}{4}$$

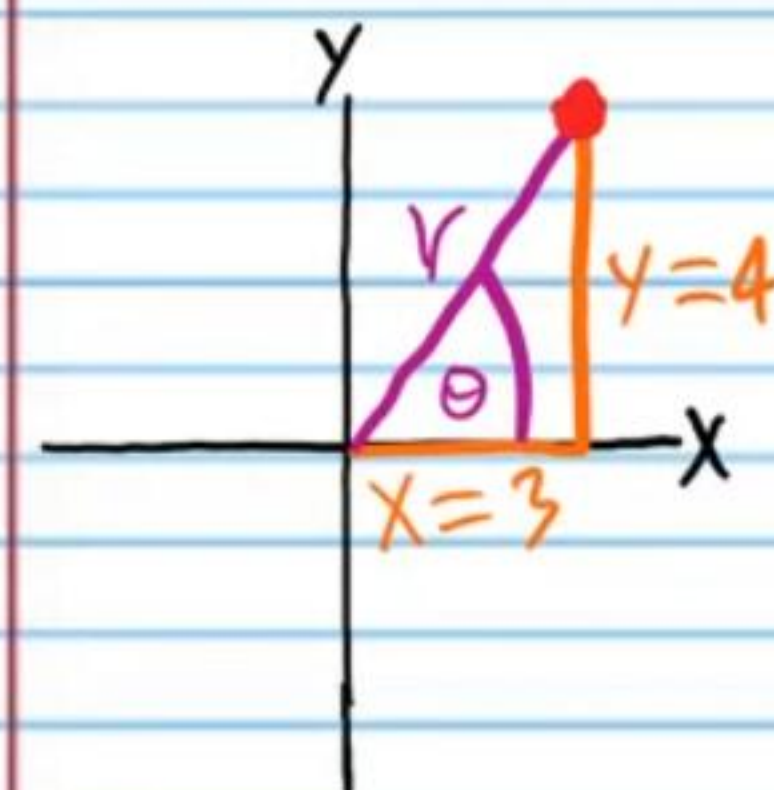
$$\tan \theta = -\frac{1}{2}$$

$$\theta = \tan^{-1}\left(-\frac{1}{2}\right)$$

$$\theta \approx -26.6^\circ$$

(15) (3, 4)

Graph



Find r.

$$x^2 + y^2 = r^2$$

$$3^2 + 4^2 = r^2$$

$$9 + 16 = r^2$$

$$25 = r^2$$

$$r = \pm 5$$

$$(5, 53.1^\circ)$$

Find theta.

$$\tan \theta = \frac{y}{x}$$

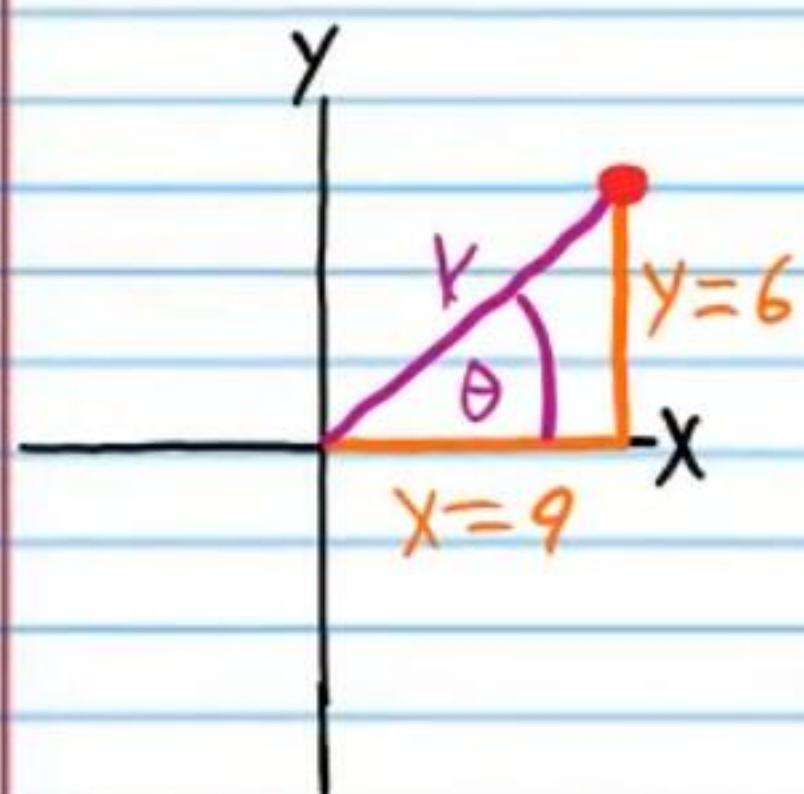
$$\tan \theta = \frac{4}{3}$$

$$\theta = \tan^{-1}\left(\frac{4}{3}\right)$$

$$\theta \approx 53.1^\circ$$

• ⑩ (9, 6)

Graph



Find r .

$$a^2 + b^2 = c^2$$

$$x^2 + y^2 = r^2$$

$$9^2 + 6^2 = r^2$$

$$81 + 36 = r^2$$

$$r^2 = 117$$

$$r = \pm\sqrt{117}$$

$$(\sqrt{117}, 33.7^\circ)$$

Find θ .

$$\tan \theta = \frac{y}{x}$$

$$\tan \theta = \frac{6}{9}$$

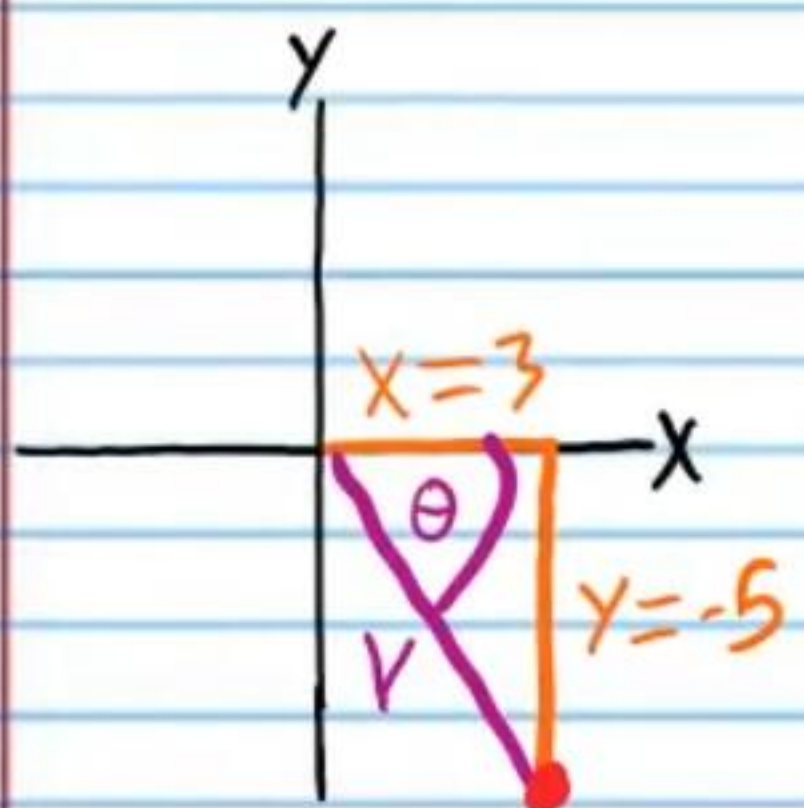
$$\tan \theta = \frac{2}{3}$$

$$\theta = \tan^{-1}\left(\frac{2}{3}\right)$$

$$\theta \approx 33.7^\circ$$

⑪ (3, -5)

Graph



Find r .

$$x^2 + y^2 = r^2$$

$$3^2 + (-5)^2 = r^2$$

$$9 + 25 = r^2$$

$$34 = r^2$$

$$r = \pm\sqrt{34}$$

$$(\sqrt{34}, -59^\circ)$$

Find θ .

$$\tan \theta = \frac{y}{x}$$

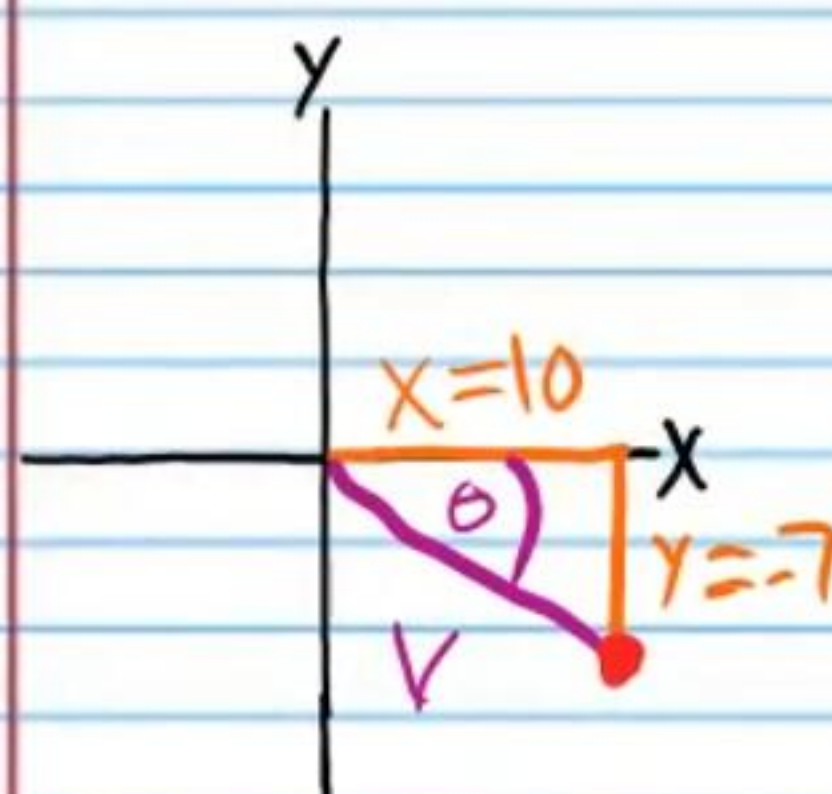
$$\tan \theta = \frac{-5}{3}$$

$$\theta = \tan^{-1}\left(-\frac{5}{3}\right)$$

$$\theta \approx -59^\circ$$

⑫ (10, -7)

Graph



Find r .

$$x^2 + y^2 = r^2$$

$$10^2 + (-7)^2 = r^2$$

$$100 + 49 = r^2$$

$$r^2 = 149$$

$$r = \pm\sqrt{149}$$

$$(\sqrt{149}, -35^\circ)$$

Find θ .

$$\tan \theta = \frac{y}{x}$$

$$\tan \theta = \frac{-7}{10}$$

$$\theta = \tan^{-1}\left(-\frac{7}{10}\right)$$

$$\theta \approx -35^\circ$$

• ⑬ (-1, -3) Graph

Find r .

$$x^2 + y^2 = r^2$$

$$(-1)^2 + (-3)^2 = r^2$$

$$1 + 9 = r^2$$

$$r^2 = 10$$

$$r = \pm\sqrt{10}$$

$$(-\sqrt{10}, 71.6^\circ)$$

Find θ .

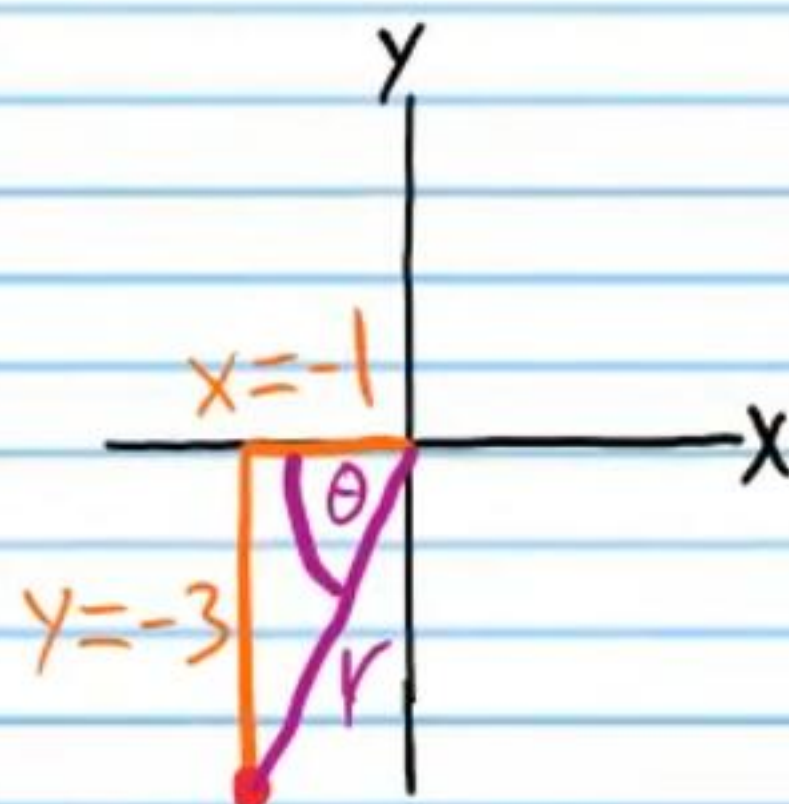
$$\tan \theta = \frac{y}{x}$$

$$\tan \theta = \frac{-3}{-1}$$

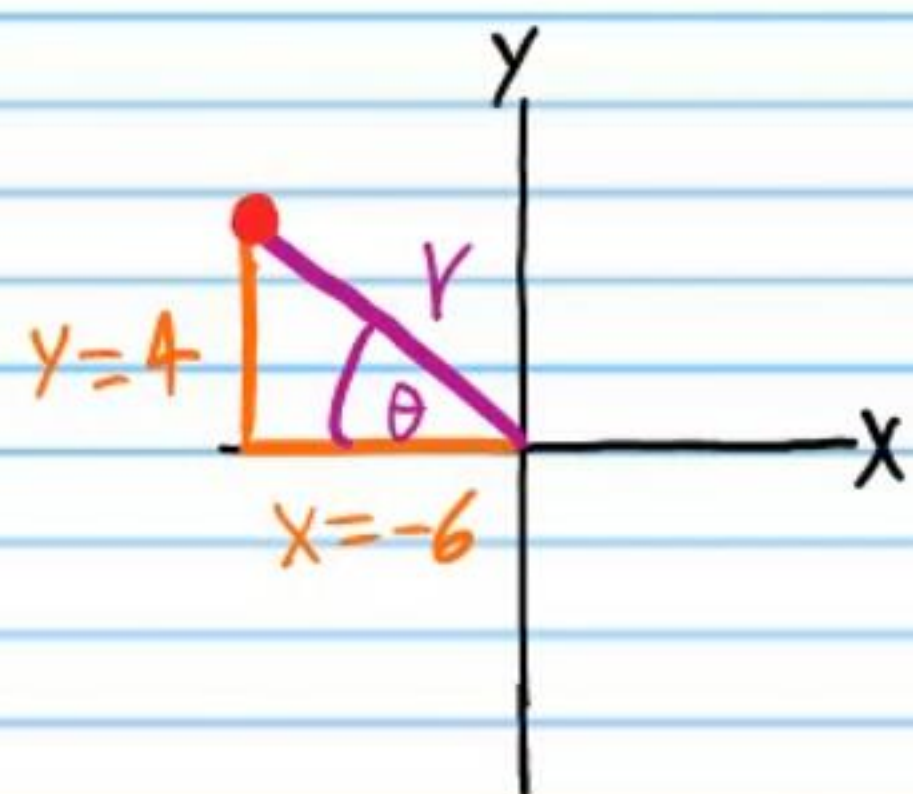
$$\tan \theta = 3$$

$$\theta = \tan^{-1} 3$$

$$\theta \approx 71.6^\circ$$



②① (-6, 4) Graph



Find r .

$$x^2 + y^2 = r^2$$

$$(-6)^2 + 4^2 = r^2$$

$$36 + 16 = r^2$$

$$52 = r^2$$

$$r = \pm\sqrt{52}$$

$$r = \pm 2\sqrt{13}$$

$$(-2\sqrt{13}, -33.7^\circ)$$

Find θ .

$$\tan\theta = \frac{y}{x}$$

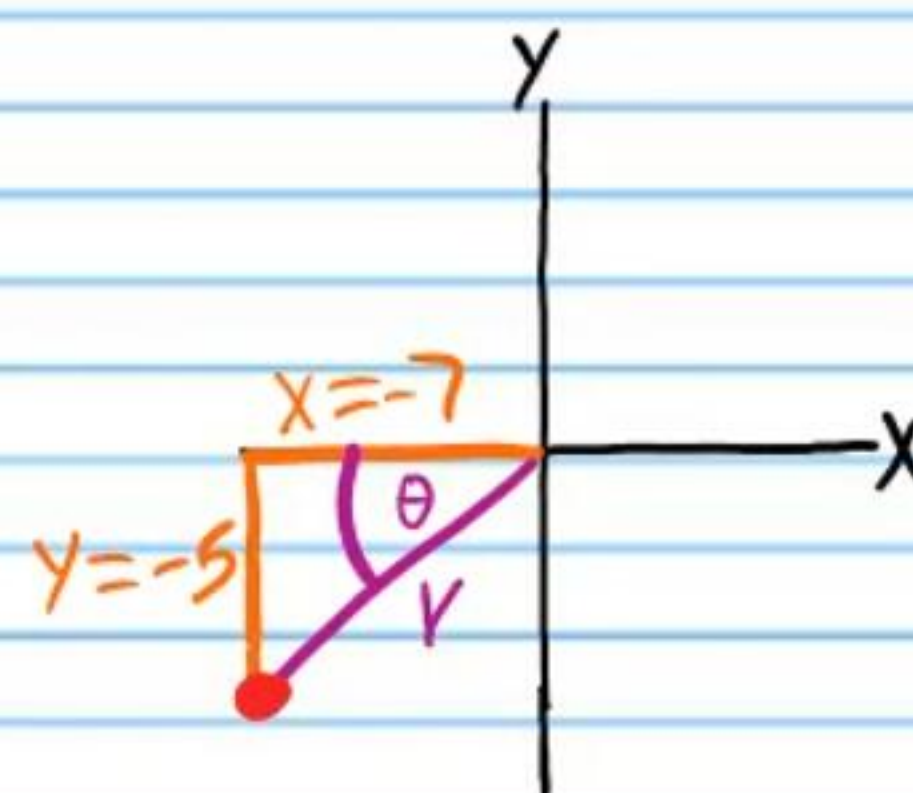
$$\tan\theta = \frac{4}{-6}$$

$$\tan\theta = -\frac{2}{3}$$

$$\theta = \tan^{-1}\left(-\frac{2}{3}\right)$$

$$\theta \approx -33.7^\circ$$

• ②① (-7, -5) Graph



Find r .

$$x^2 + y^2 = r^2$$

$$(-7)^2 + (-5)^2 = r^2$$

$$49 + 25 = r^2$$

$$r^2 = 74$$

$$r = \pm\sqrt{74}$$

$$(-\sqrt{74}, 35.5^\circ)$$

Find θ .

$$\tan\theta = \frac{y}{x}$$

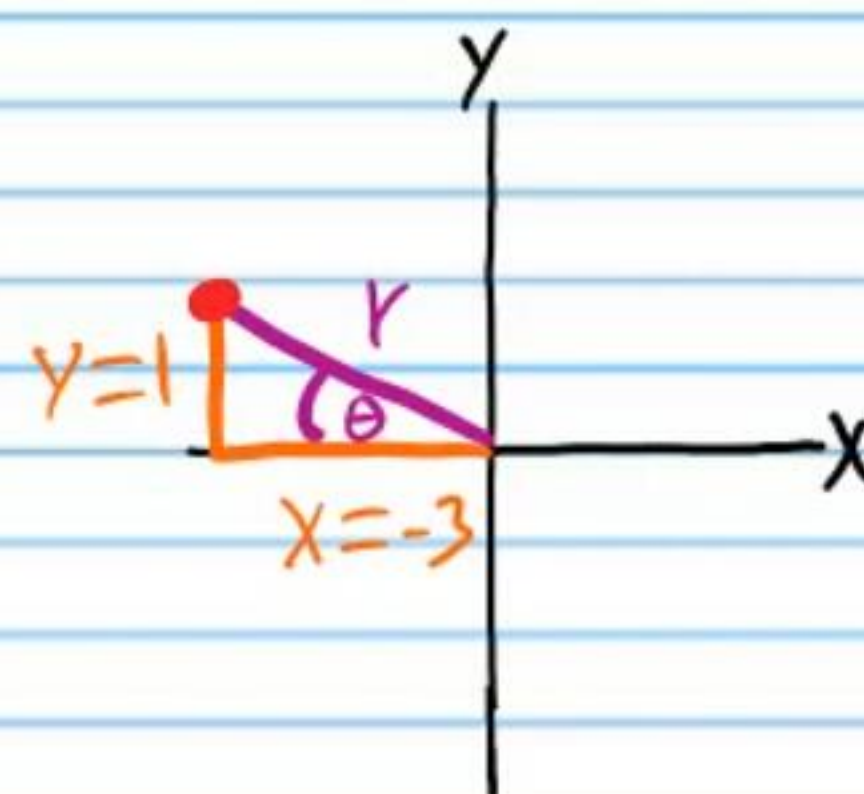
$$\tan\theta = \frac{-5}{-7}$$

$$\tan\theta = \frac{5}{7}$$

$$\theta = \tan^{-1}\left(\frac{5}{7}\right)$$

$$\theta \approx 35.5^\circ$$

②② (-3, 1) Graph



Find r .

$$x^2 + y^2 = r^2$$

$$(-3)^2 + 1^2 = r^2$$

$$9 + 1 = r^2$$

$$r^2 = 10$$

$$r = \pm\sqrt{10}$$

$$(-\sqrt{10}, -18.4^\circ)$$

Find θ .

$$\tan\theta = \frac{y}{x}$$

$$\tan\theta = \frac{1}{-3}$$

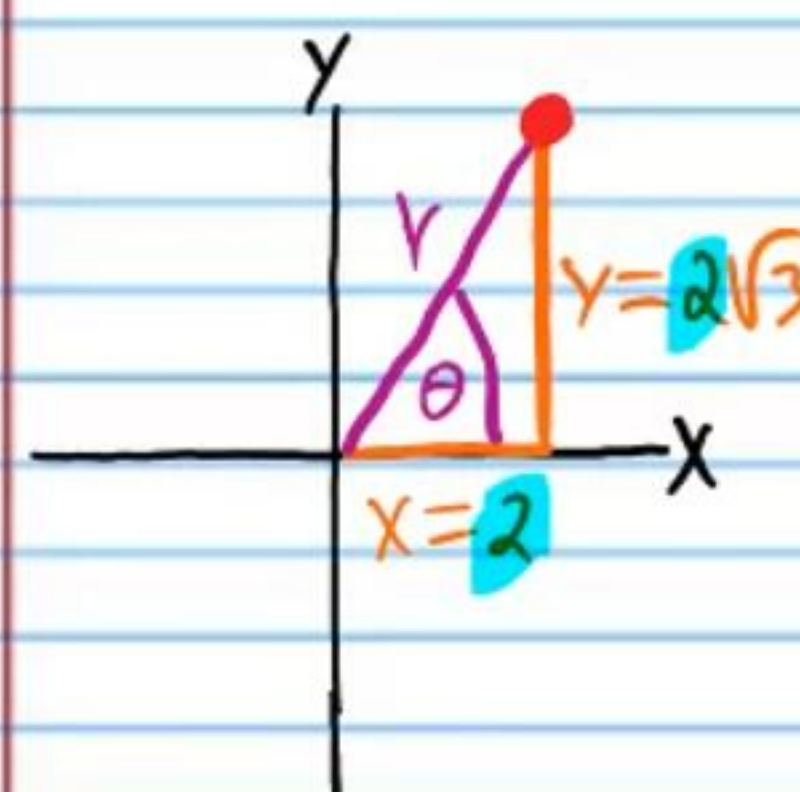
$$\theta = \tan^{-1}\left(-\frac{1}{3}\right)$$

$$\theta \approx -18.4^\circ$$

Do not use a calculator for the problems below.

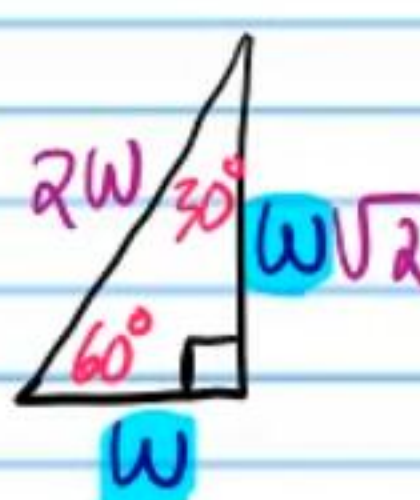
• ②③ (2, 2√3)

Graph



Find r .

Find θ .



$$r = 2w$$

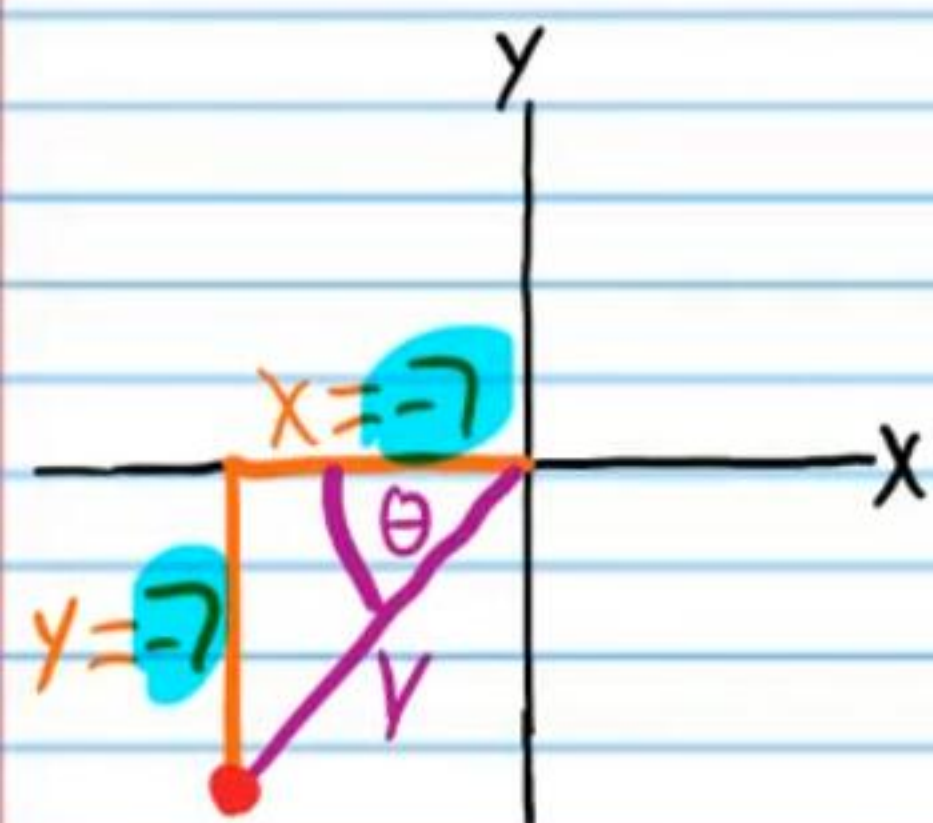
$$r = 2(2)$$

$$r = 4$$

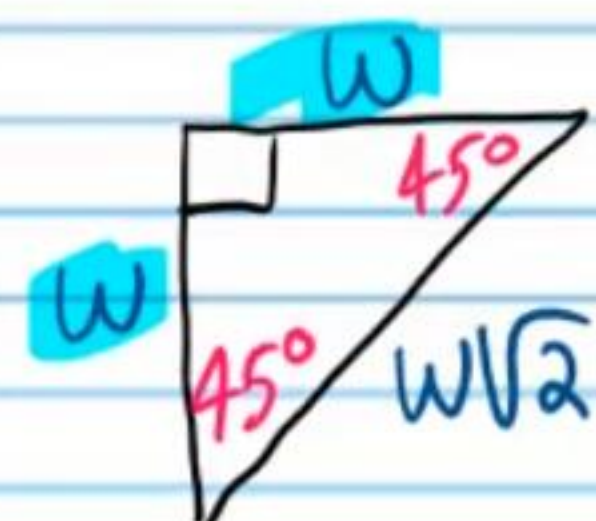
$$(4, 60^\circ)$$

②④ $(-7, -7)$

Graph



Find r .



$$r = w\sqrt{2}$$

$$r = -7\sqrt{2}$$

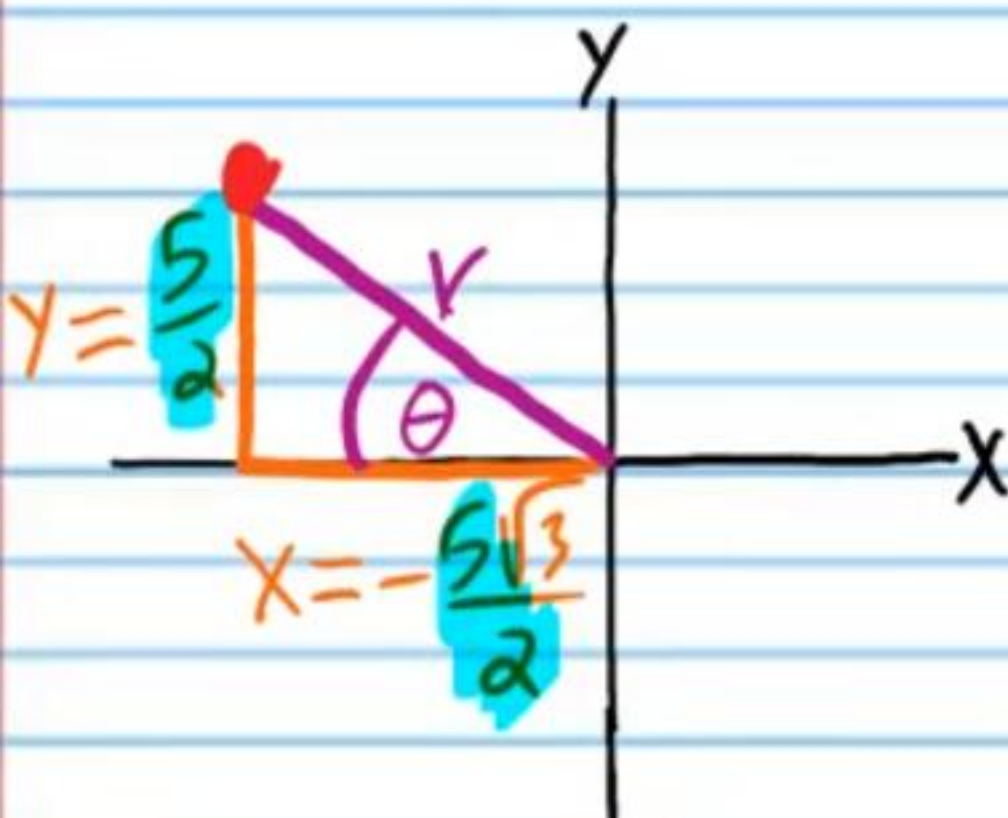
Find θ .

$$\theta = 45^\circ$$

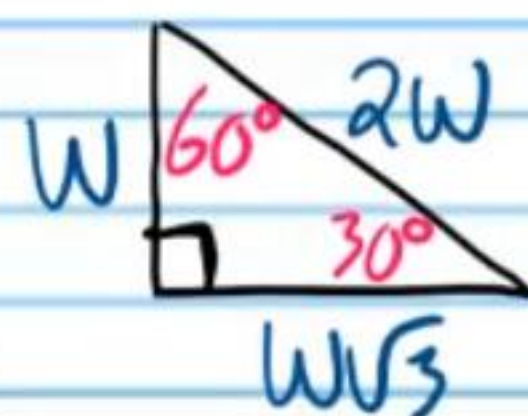
$$(-7\sqrt{2}, 45^\circ)$$

②⑤ $(-\frac{5\sqrt{3}}{2}, \frac{5}{2})$

Graph



Find r .



$$r = 2w$$

$$r = 2(\frac{5}{2})$$

$$r = 5$$

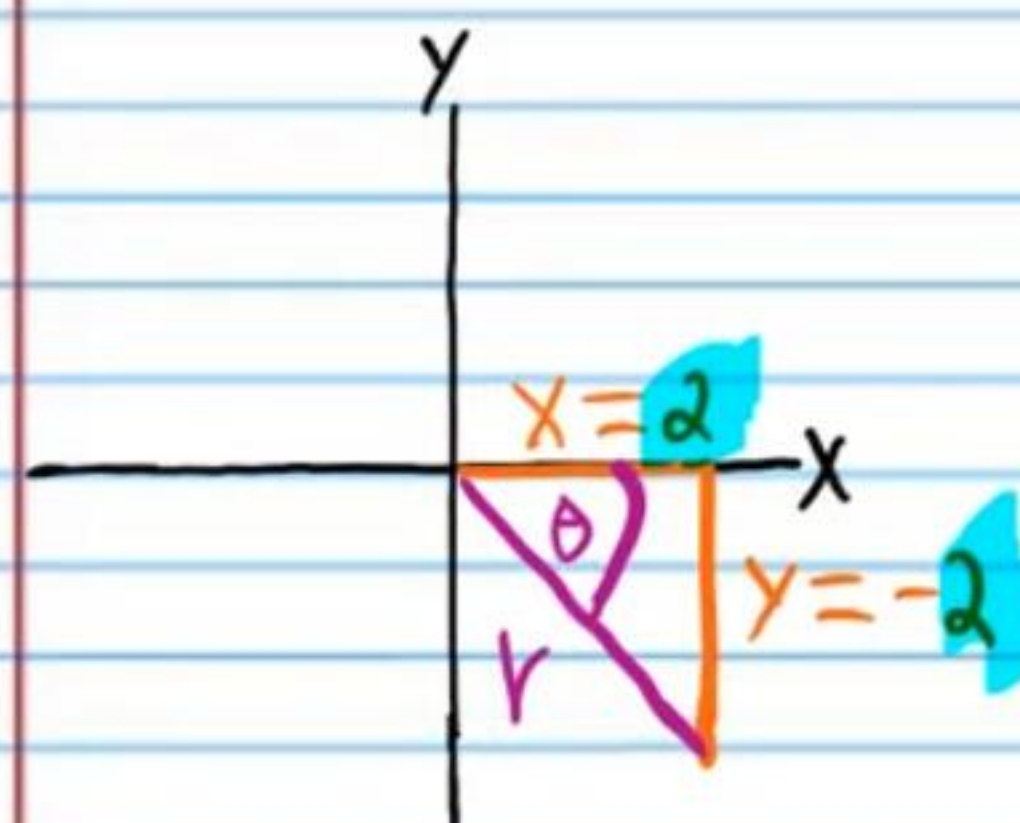
Find θ .

$$\theta = 30^\circ$$

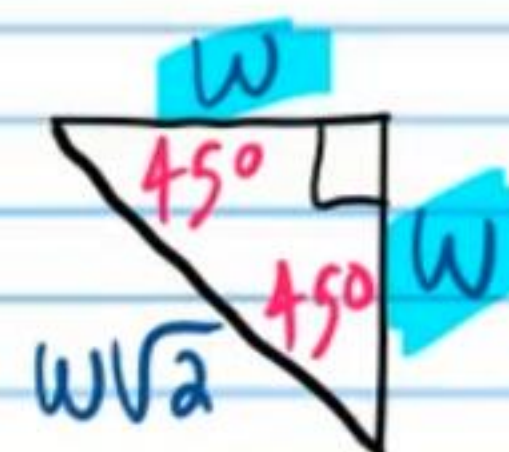
$$(-5, -30^\circ)$$

②⑥ $(2, -2)$

Graph



Find r .



$$r = w\sqrt{2}$$

$$r = 2\sqrt{2}$$

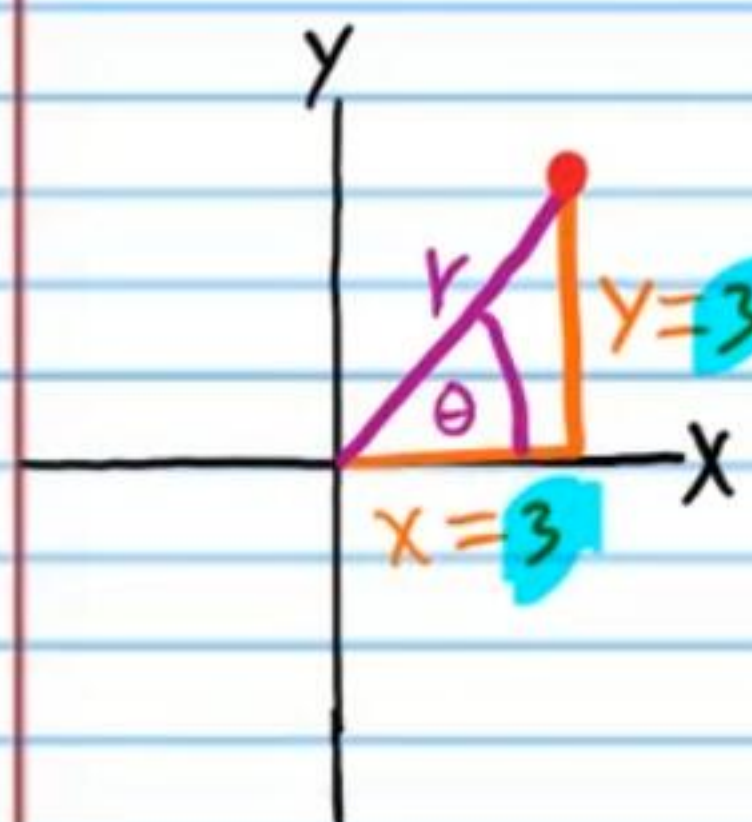
Find θ .

$$\theta = 45^\circ$$

$$(2\sqrt{2}, -45^\circ)$$

• ②⑦ $(3, 3)$

Graph



Find r .



$$r = w\sqrt{2}$$

$$r = 3\sqrt{2}$$

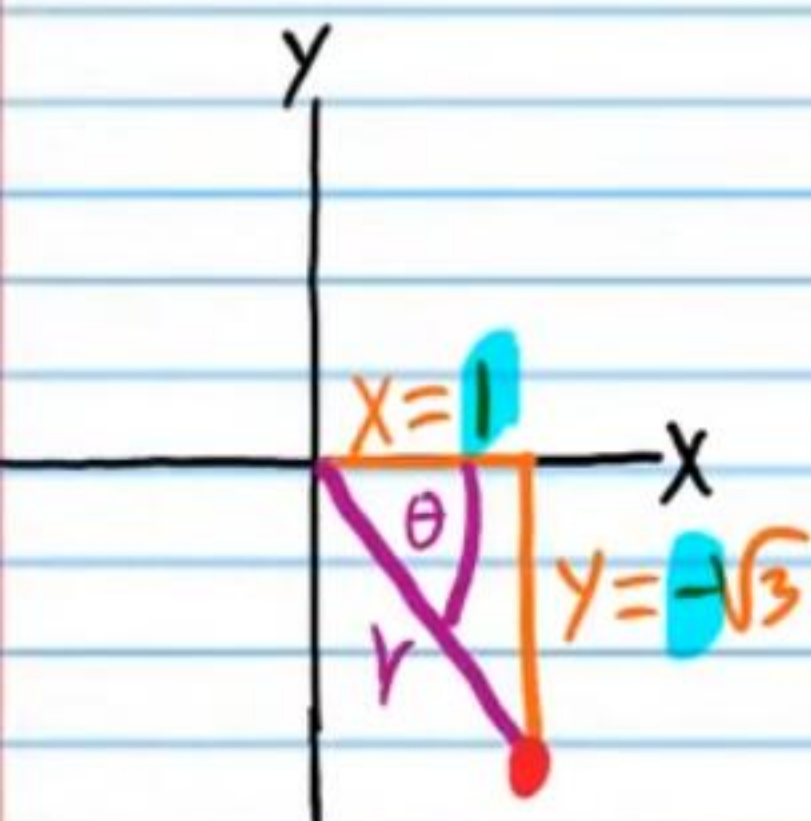
Find θ .

$$\theta = 45^\circ$$

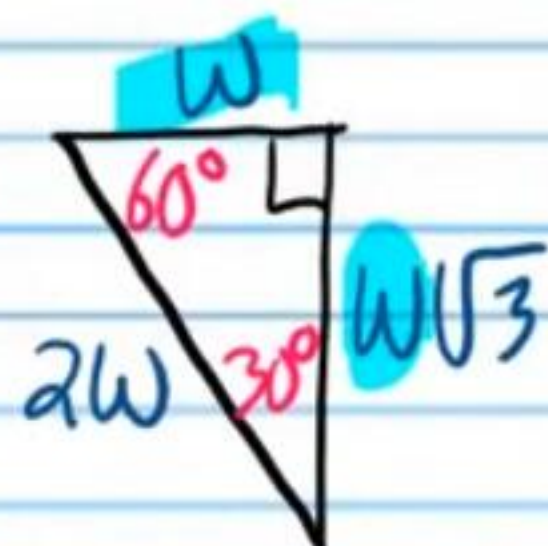
$$(3\sqrt{2}, 45^\circ)$$

②⑧ $(1, -\sqrt{3})$

Graph



Find r .



$$r = 2w$$

$$r = 2(1)$$

$$r = 2$$

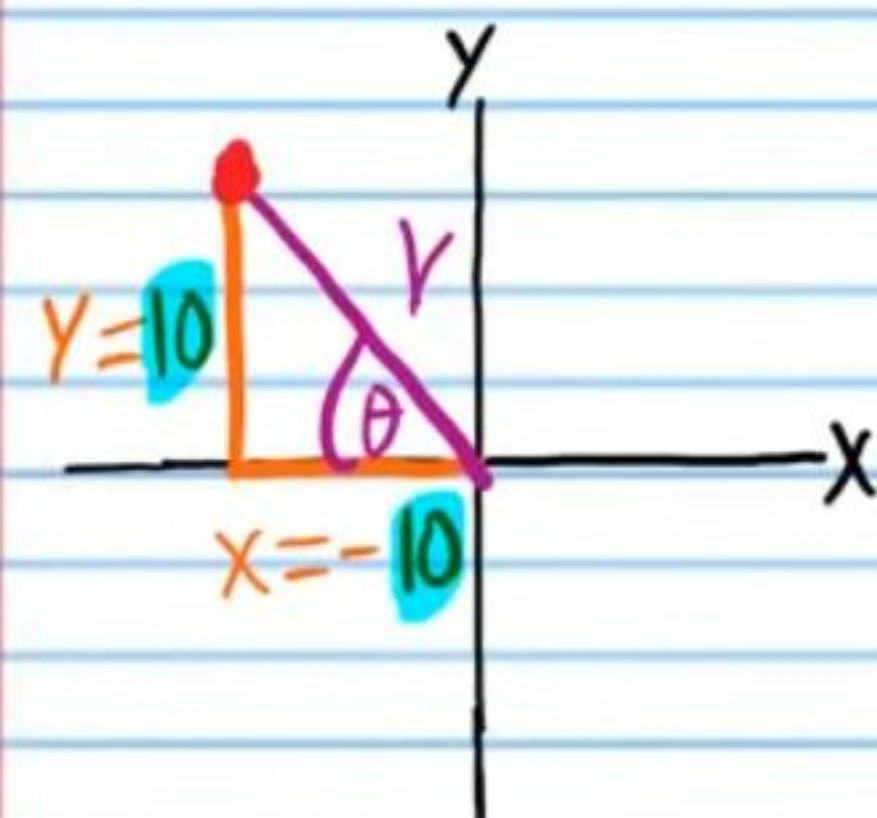
Find θ .

$$\theta = 60^\circ$$

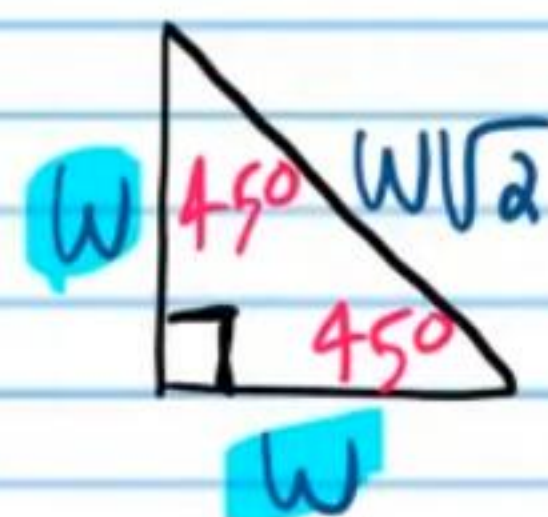
$$(2, -60^\circ)$$

②⑨ $(-10, 10)$

Graph



Find r .



$$r = w\sqrt{2}$$

$$r = 10\sqrt{2}$$

Find θ .

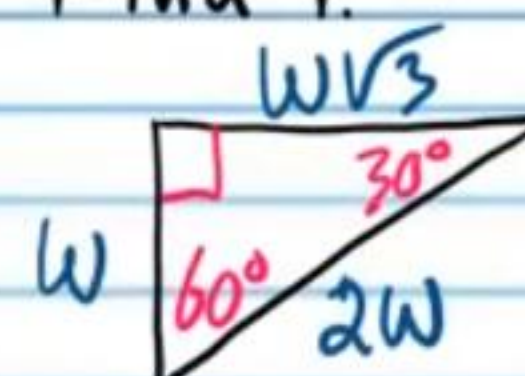
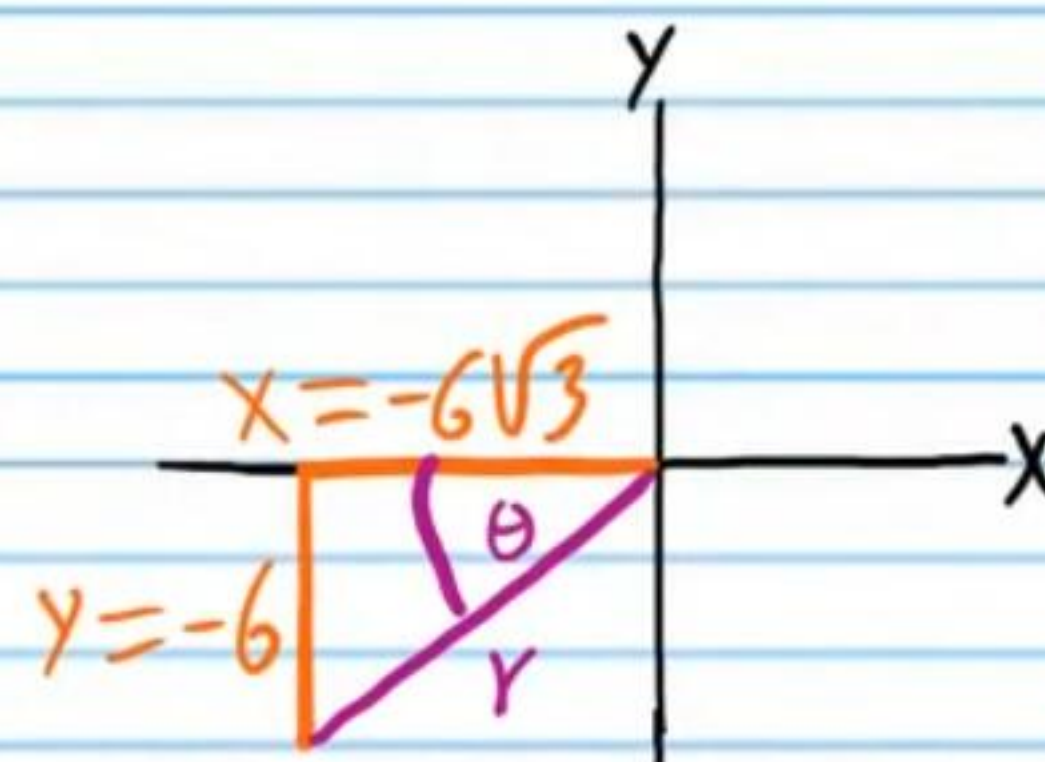
$$\theta = 45^\circ$$

$$(-10\sqrt{2}, -45^\circ)$$

③⑩ $(-6\sqrt{3}, -6)$ Graph

Find r .

Find θ .



$$r = 2w$$

$$r = 2(6)$$

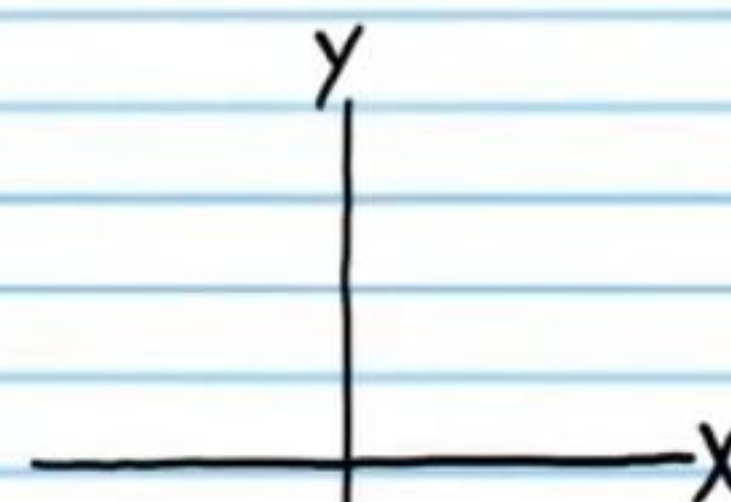
$$r = 12$$

$$(-12, 30^\circ)$$

• ③⑪ $(0, -1\frac{1}{3})$ Graph

Find r .

Find θ .



$$r = 1\frac{1}{3}$$

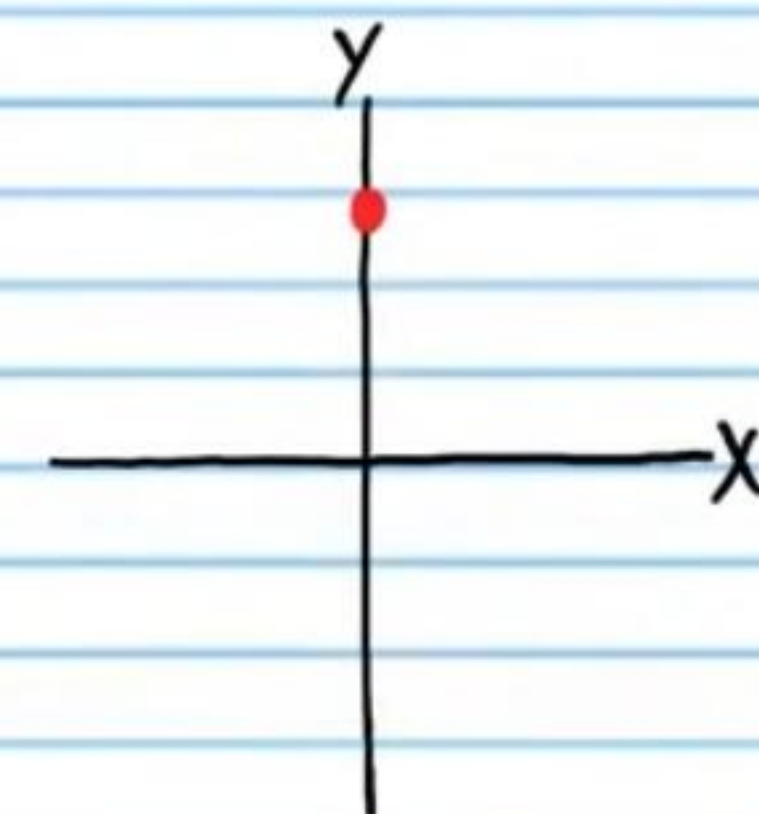
$$\theta = -90^\circ$$

$$(1\frac{1}{3}, -90^\circ)$$

③⑫ $(0, 4)$ Graph

Find r .

Find θ .



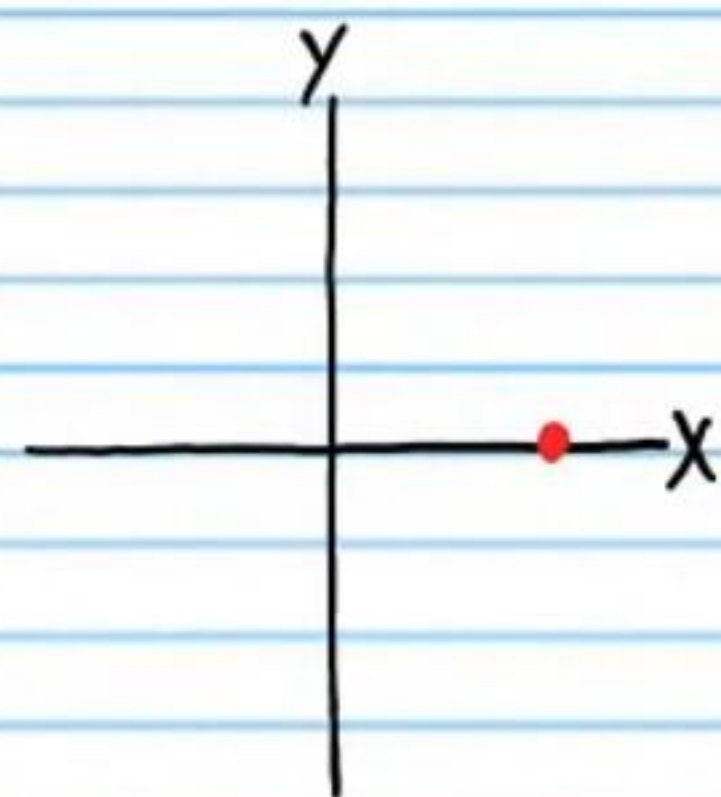
$$r = 4$$

$$\theta = 90^\circ$$

$$(4, 90^\circ)$$

③③ (3,0) Graph Find r. Find θ .

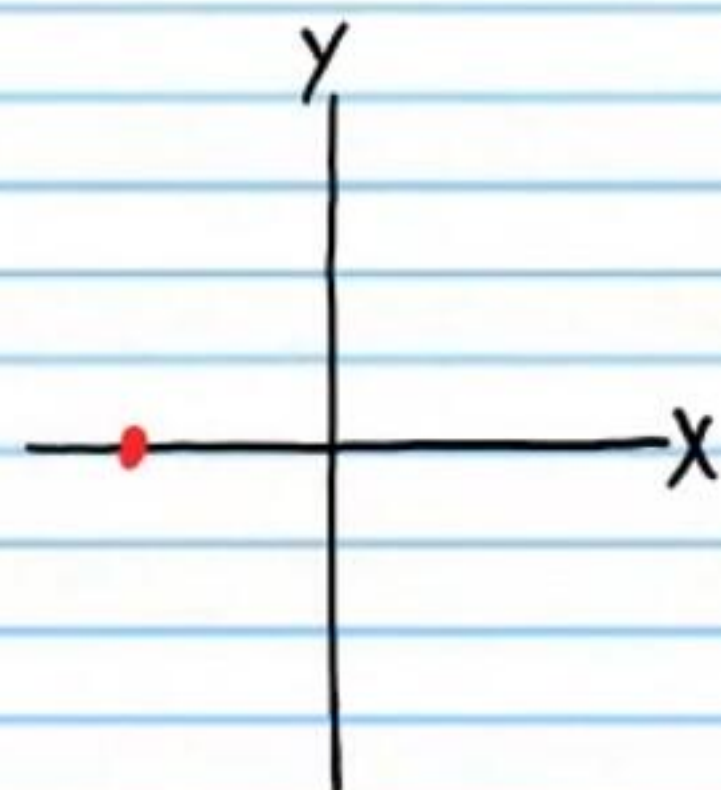
$r = 3$ $\theta = 0^\circ$



$(3, 0^\circ)$

③④ (-2,0) Graph Find r. Find θ .

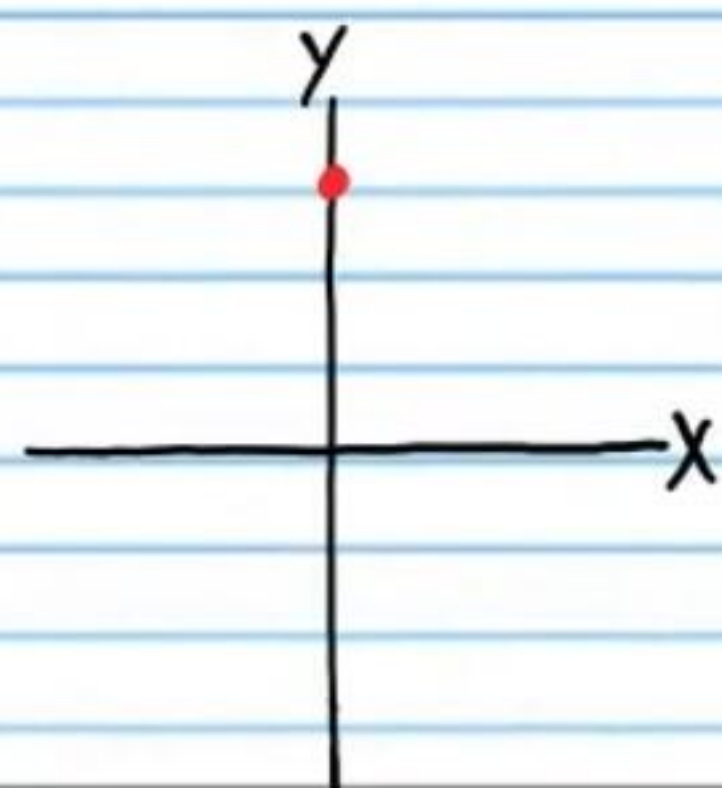
$r = -2$ $\theta = 0^\circ$



$(-2, 0^\circ)$

• ③⑤ (0,9) Graph Find r. Find θ .

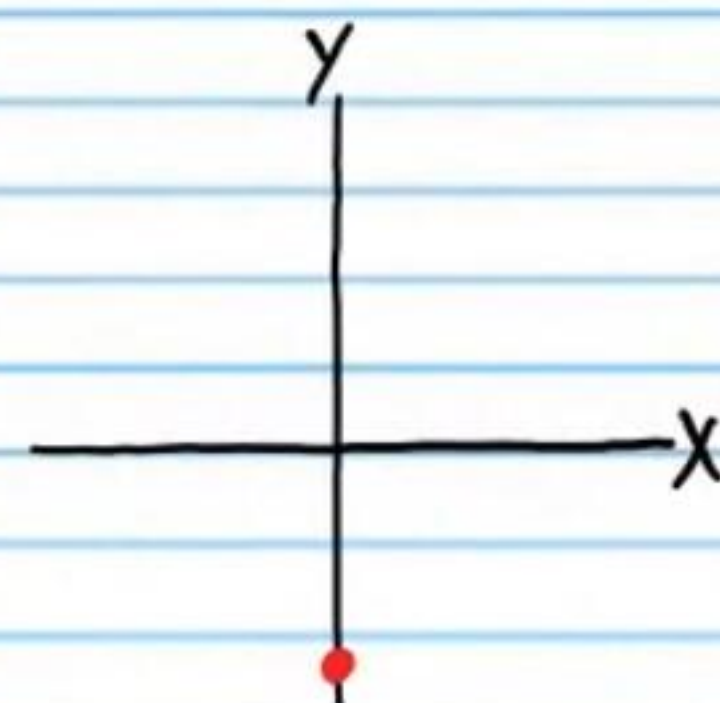
$r = 9$ $\theta = 90^\circ$



$(9, 90^\circ)$

③⑥ (0,-10) Graph Find r. Find θ .

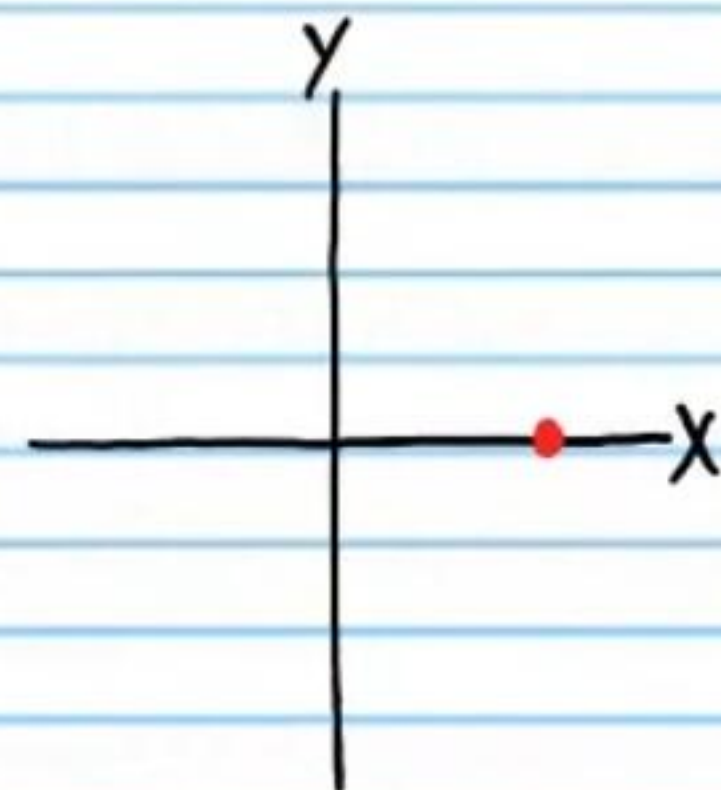
$r = 10$ $\theta = -90^\circ$



$(10, -90^\circ)$

③⑦ (1,0) Graph Find r. Find θ .

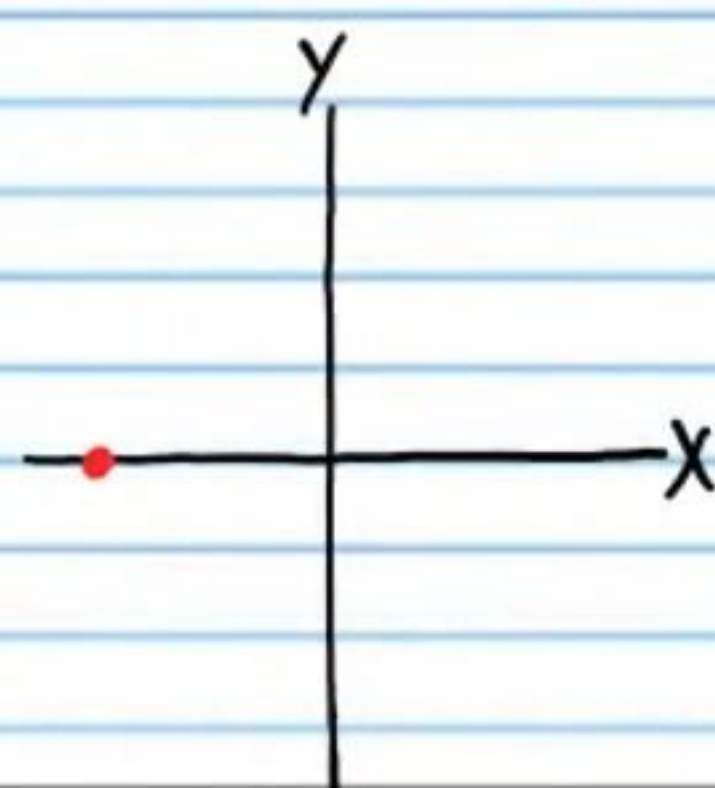
$r = 1$ $\theta = 0^\circ$



$(1, 0^\circ)$

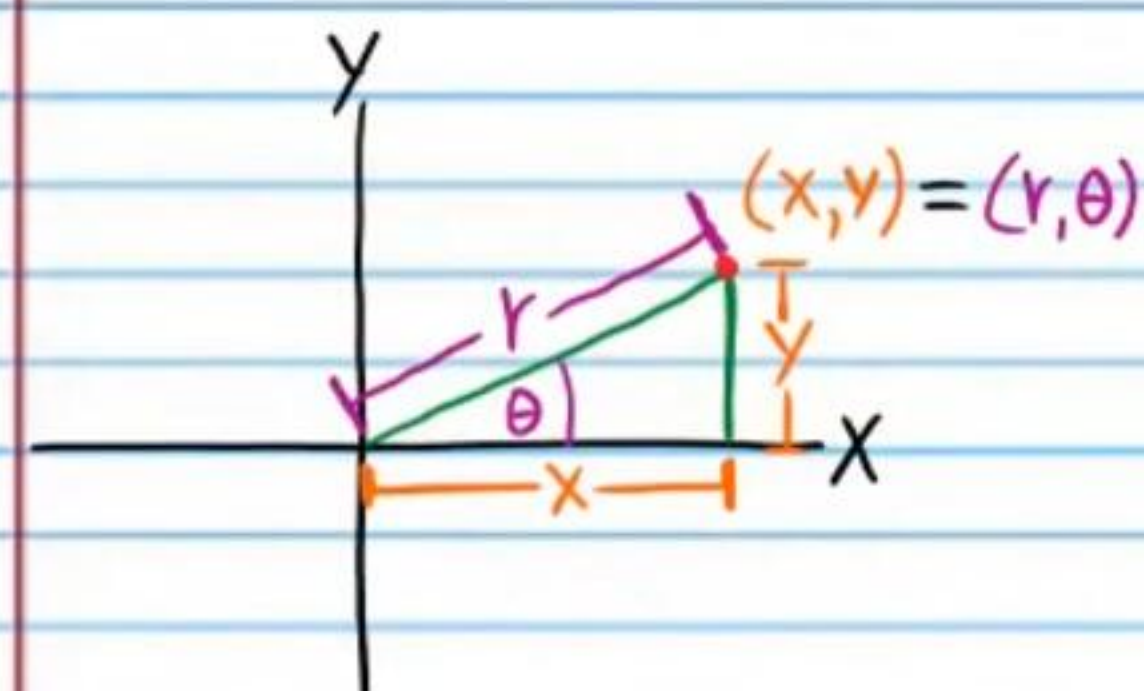
③⑧ (-5,0) Graph Find r. Find θ .

$r = -5$ $\theta = 0^\circ$



$(-5, 0^\circ)$

Converting from Polar Coordinates to Rectangular Coordinates



$$\cos \theta = \frac{x}{r}$$

$$\sin \theta = \frac{y}{r}$$

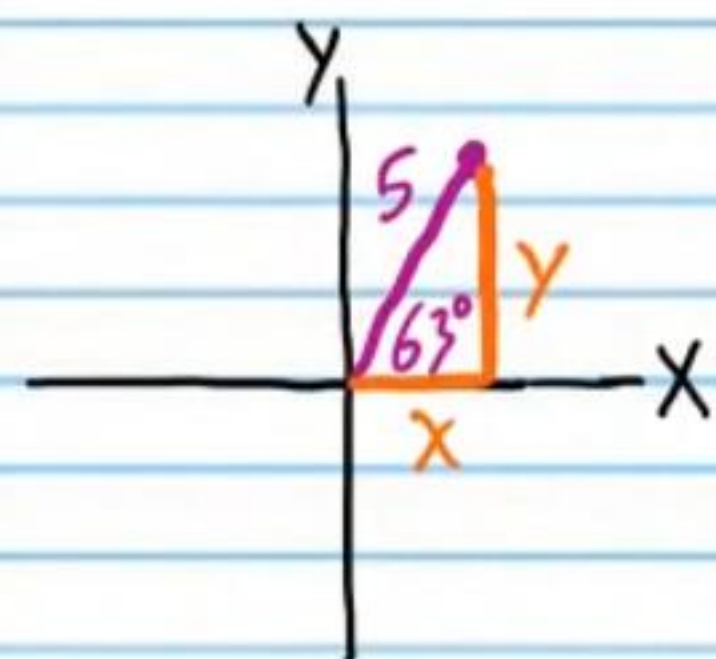
$$x = r \cos \theta$$

$$y = r \sin \theta$$

Convert the following polar coordinates to rectangular coordinates.
Round to the nearest hundredth.

• ③⑨ $(5, 63^\circ)$

Graph



Find x .

$$\cos \theta = \frac{x}{r}$$

Find y .

$$\sin \theta = \frac{y}{r}$$

$$\cos 63^\circ = \frac{x}{5}$$

$$\sin 63^\circ = \frac{y}{5}$$

$$5 \cos 63^\circ = x$$

$$5 \sin 63^\circ = y$$

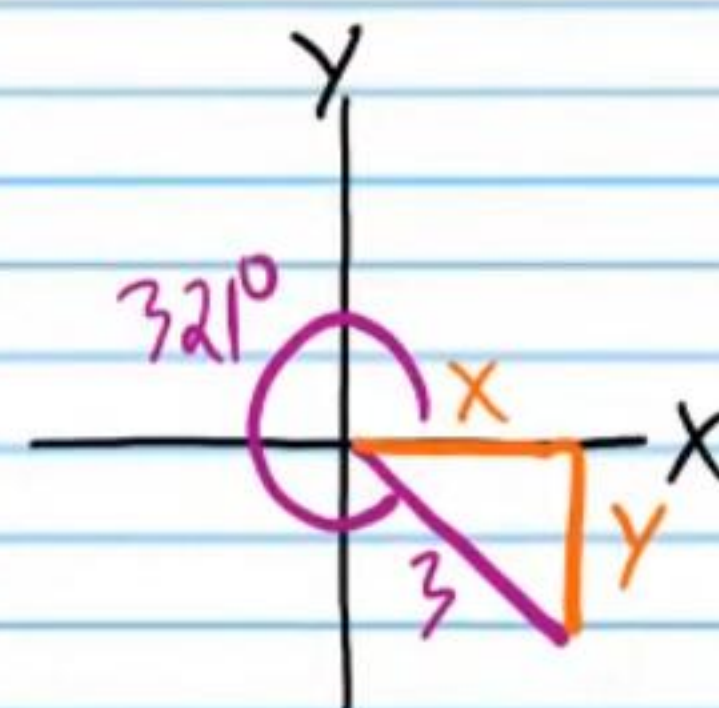
$$x \approx 2.27$$

$$y \approx 4.46$$

$$\boxed{(2.27, 4.46)}$$

④⑩ $(3, 321^\circ)$

Graph



Find x .

$$x = r \cos \theta$$

$$x = 3 \cos 321^\circ$$

$$x \approx 2.33$$

Find y .

$$y = r \sin \theta$$

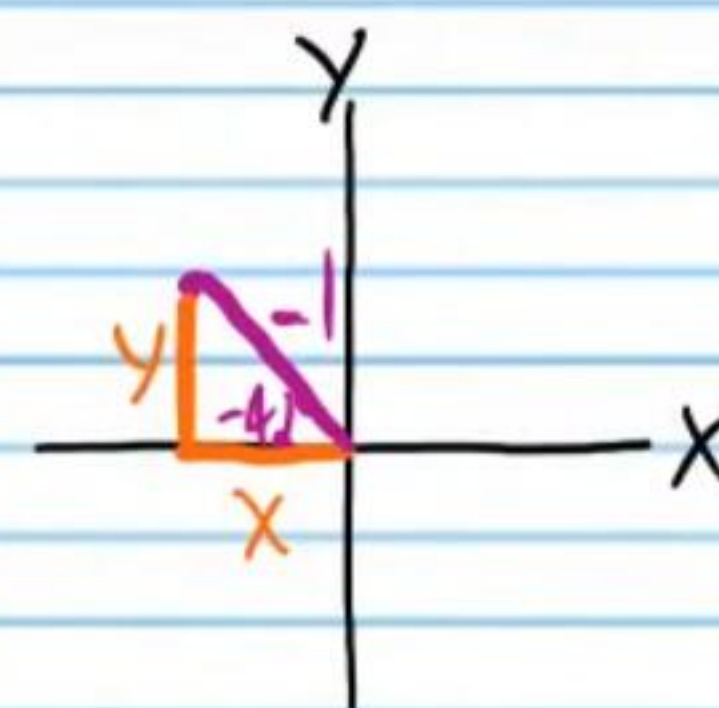
$$y = 3 \sin 321^\circ$$

$$y \approx -1.89$$

$$\boxed{(2.33, -1.89)}$$

④⑪ $(-1, -42^\circ)$

Graph



Find x .

$$x = r \cos \theta$$

$$x = -1 \cos(-42^\circ)$$

$$x \approx -0.74$$

Find y .

$$y = r \sin \theta$$

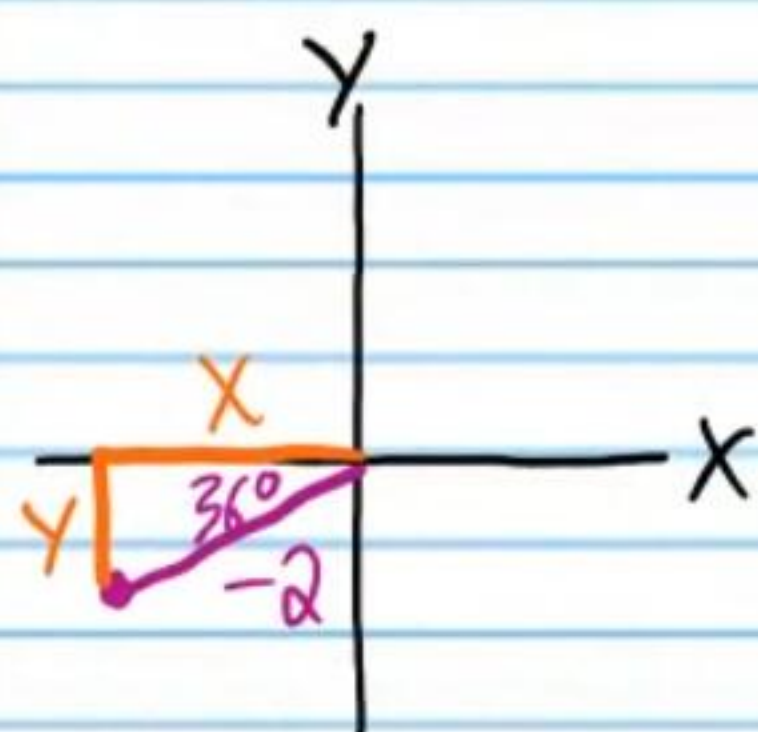
$$y = -1 \sin(-42^\circ)$$

$$y \approx 0.67$$

$$\boxed{(-0.74, 0.67)}$$

42) $(-2, 36^\circ)$

Graph



Find x .

$$x = r \cos \theta$$

$$x = -2 \cos 36^\circ$$

$$x \approx -1.62$$

Find y .

$$y = r \sin \theta$$

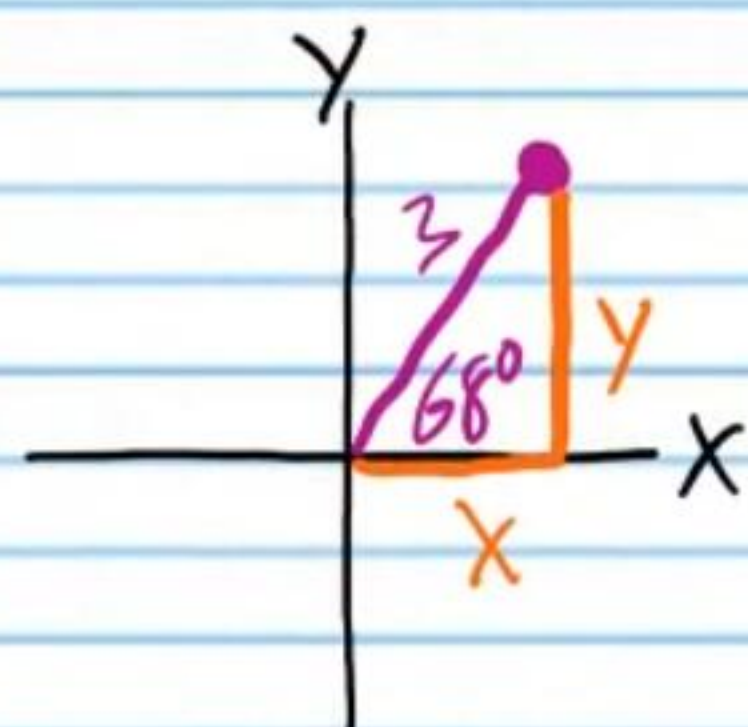
$$y = -2 \sin 36^\circ$$

$$y \approx -1.18$$

$$\boxed{(-1.62, -1.18)}$$

43) $(3, 68^\circ)$

Graph



Find x .

$$x = r \cos \theta$$

$$x = 3 \cos 68^\circ$$

$$x \approx 1.12$$

Find y .

$$y = r \sin \theta$$

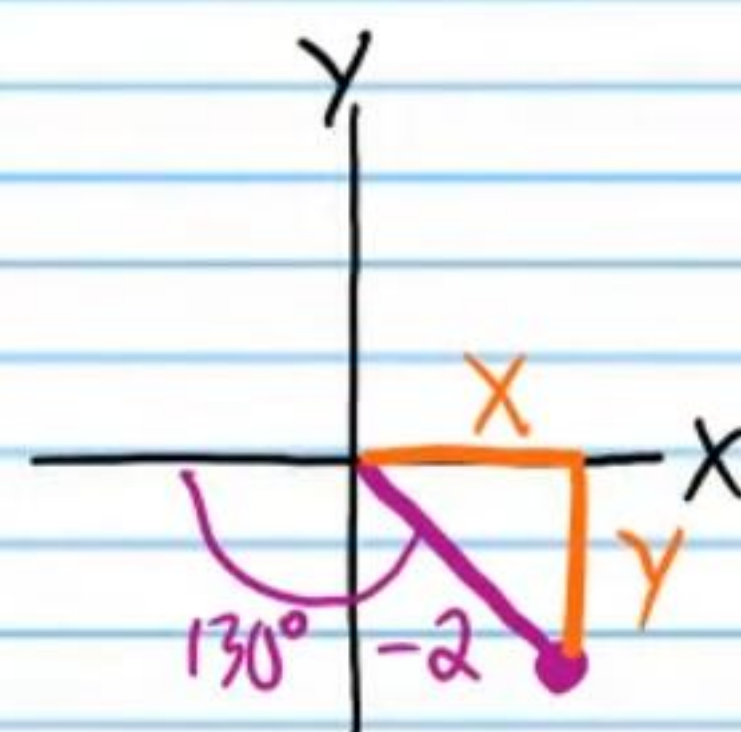
$$y = 3 \sin 68^\circ$$

$$y \approx 2.78$$

$$\boxed{(1.12, 2.78)}$$

44) $(-2, 130^\circ)$

Graph



Find x .

$$x = r \cos \theta$$

$$x = -2 \cos 130^\circ$$

$$x \approx 1.29$$

Find y .

$$y = r \sin \theta$$

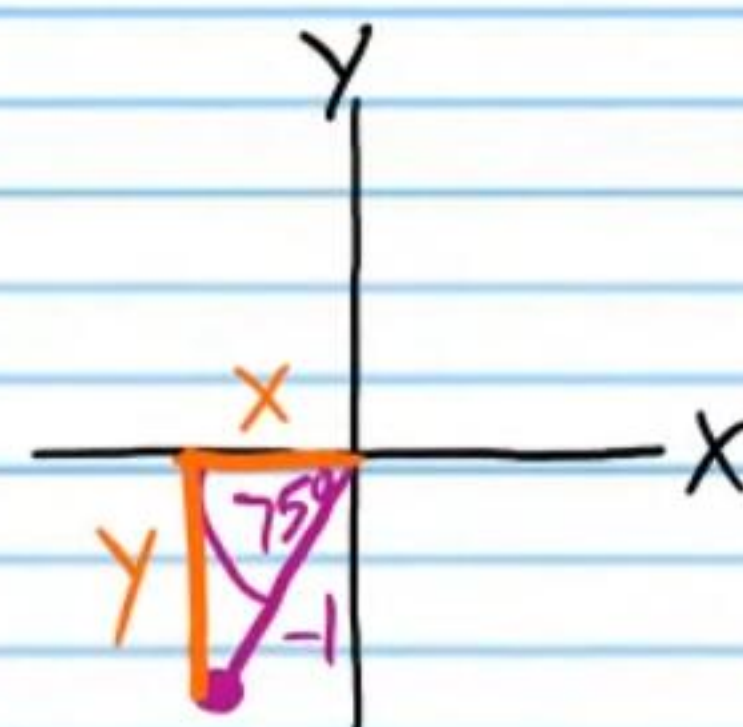
$$y = -2 \sin 130^\circ$$

$$y \approx -1.53$$

$$\boxed{(1.29, -1.53)}$$

45) $(-1, 75^\circ)$

Graph



Find x .

$$x = r \cos \theta$$

$$x = -\cos 75^\circ$$

$$x \approx -0.26$$

Find y .

$$y = r \sin \theta$$

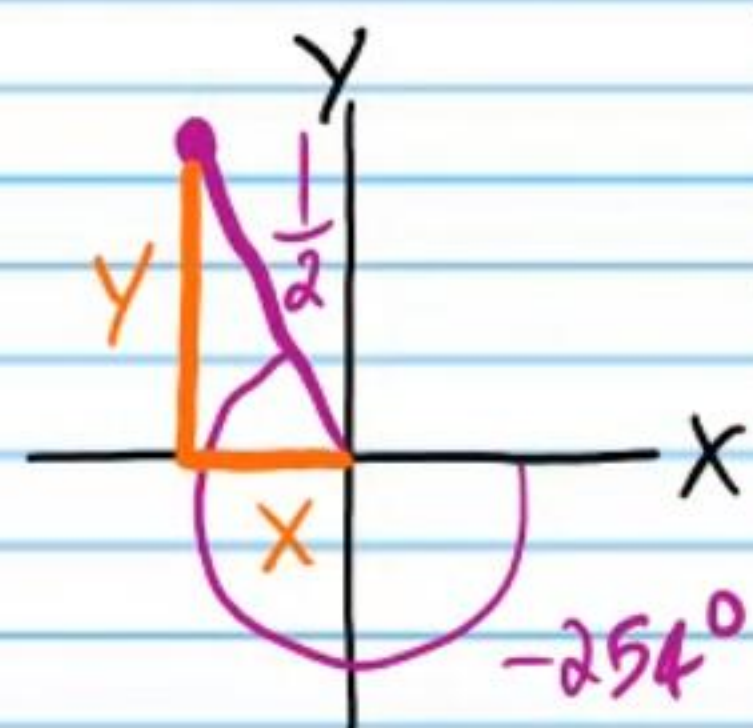
$$y = -\sin 75^\circ$$

$$y \approx -0.97$$

$$\boxed{(-0.26, -0.97)}$$

④⑥ $(\frac{1}{2}, -254^\circ)$

Graph



Find x .

$$x = r \cos \theta$$

$$x = \frac{1}{2} \cos(-254^\circ)$$

$$x \approx -0.14$$

Find y .

$$y = r \sin \theta$$

$$y = \frac{1}{2} \sin(-254^\circ)$$

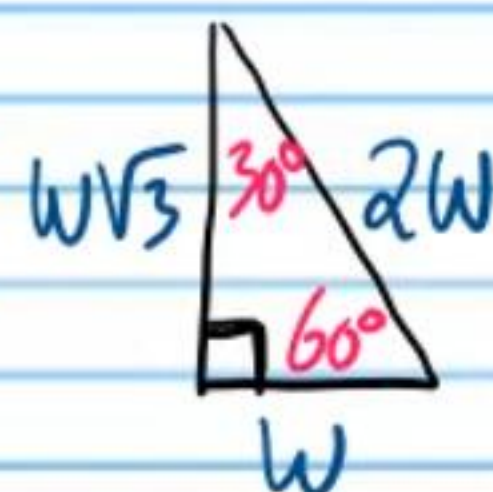
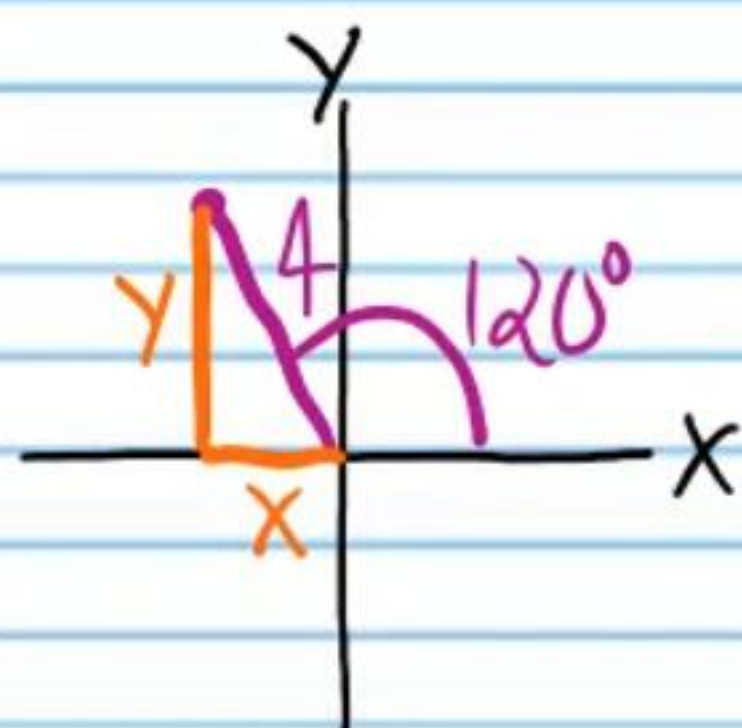
$$y \approx 0.48$$

$$\boxed{(-0.14, 0.48)}$$

Do not use a calculator for the problems below.

④⑦ $(4, 120^\circ)$

Graph



$$r = 2W$$

$$4 = 2W$$

$$W = 2$$

Find x . Find y .

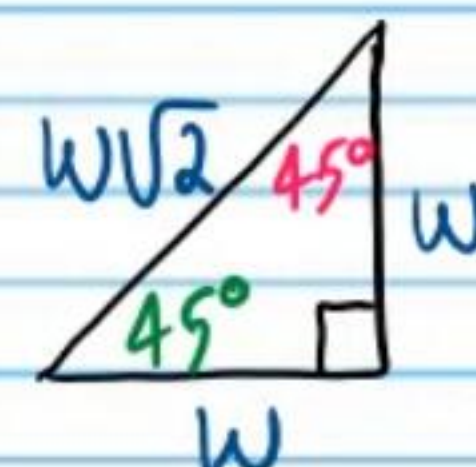
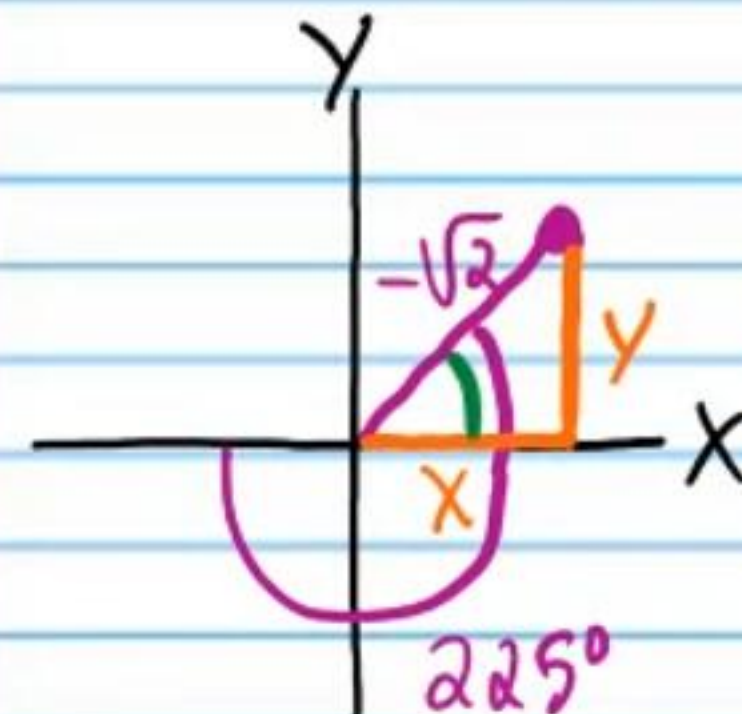
$$x = -W \quad y = W\sqrt{3}$$

$$x = -2 \quad y = 2\sqrt{3}$$

$$\boxed{(-2, 2\sqrt{3})}$$

④⑧ $(-\sqrt{2}, 225^\circ)$

Graph



$$r = W\sqrt{2}$$

$$-\sqrt{2} = W\sqrt{2}$$

$$-1 = W$$

$$W = 1$$

Find x . Find y .

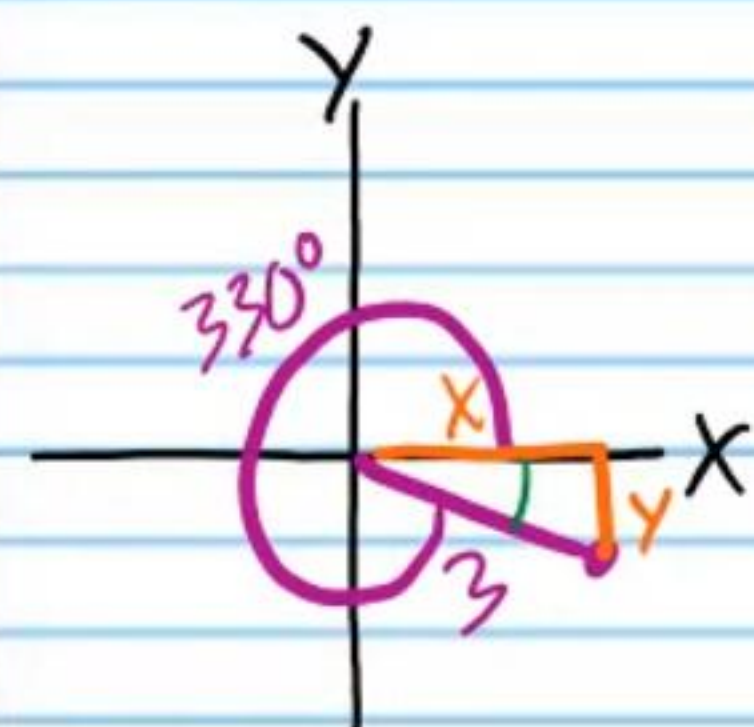
$$x = W \quad y = W$$

$$x = 1 \quad y = 1$$

$$\boxed{(1, 1)}$$

④⑨ $(3, 330^\circ)$

Graph



$$r = 2W$$

$$3 = 2W$$

$$W = \frac{3}{2}$$

Find x . Find y .

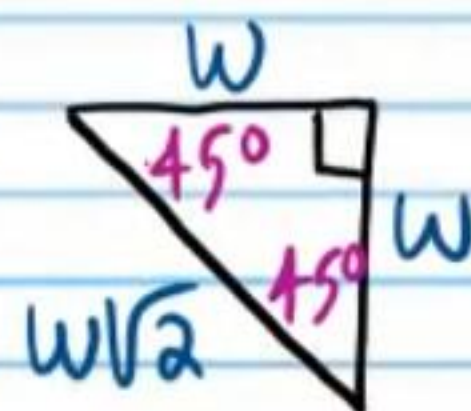
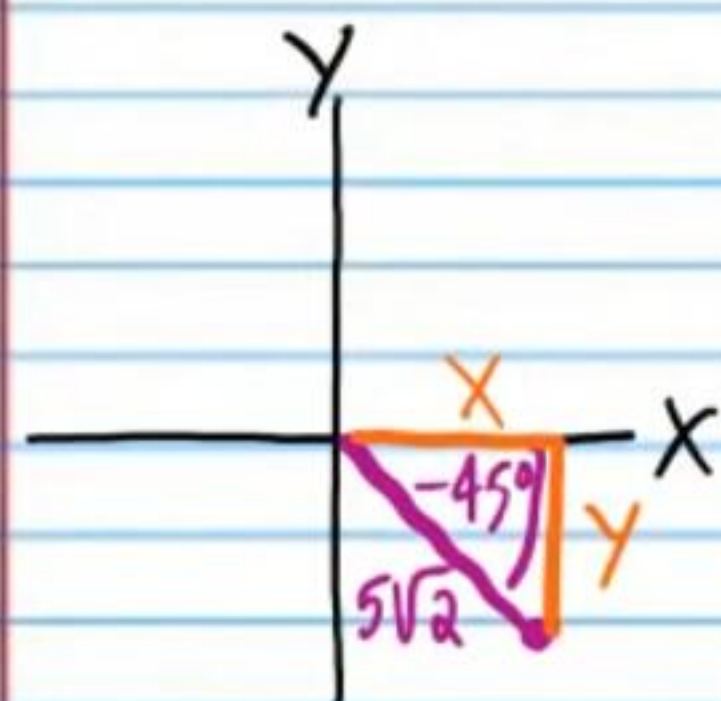
$$x = W\sqrt{3} \quad y = -W$$

$$x = \frac{3}{2}\sqrt{3} \quad y = -\frac{3}{2}$$

$$\boxed{\left(\frac{3}{2}\sqrt{3}, -\frac{3}{2}\right)}$$

⑤① $(5\sqrt{2}, -45^\circ)$

Graph



$$r = w\sqrt{2}$$

$$5\sqrt{2} = w\sqrt{2}$$

$$w = 5$$

Find X. Find y.

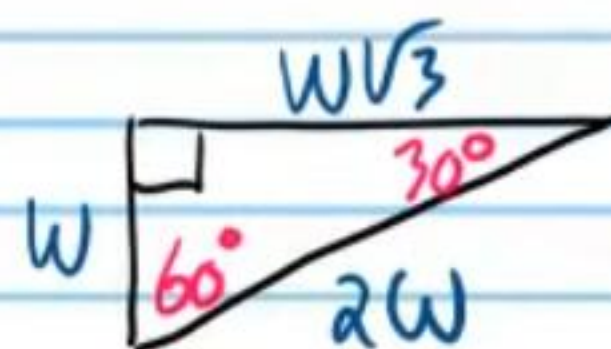
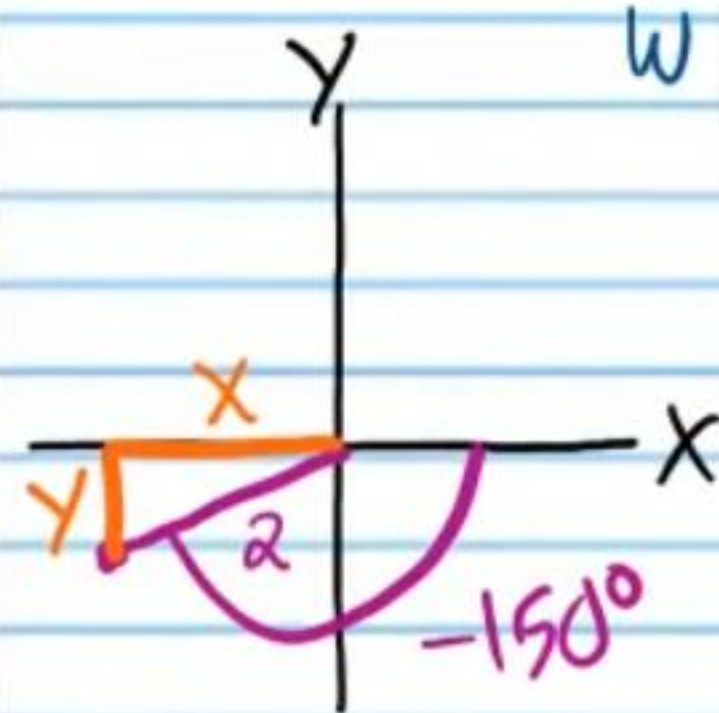
$$x = w \quad y = -w$$

$$x = 5 \quad y = -5$$

$$(5, -5)$$

• ⑤② $(2, -150^\circ)$

Graph



$$r = 2w$$

$$2 = 2w$$

$$w = 1$$

Find X. Find y.

$$x = -w\sqrt{3} \quad y = -w$$

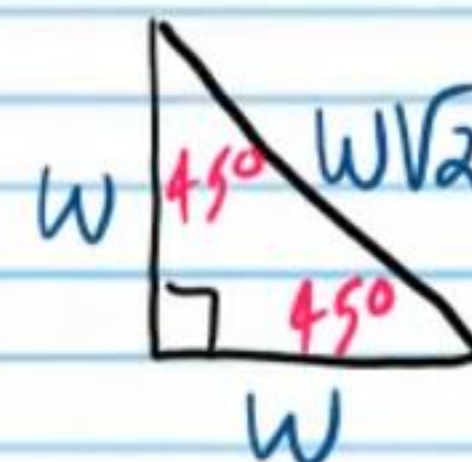
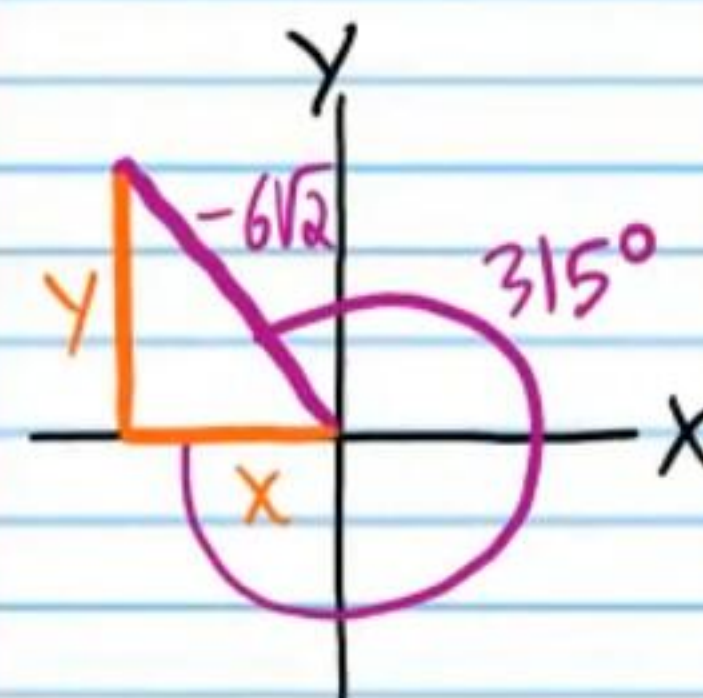
$$x = -(1)\sqrt{3} \quad y = -1$$

$$x = -\sqrt{3}$$

$$(-\sqrt{3}, -1)$$

⑤③ $(-6\sqrt{2}, 315^\circ)$

Graph



$$r = w\sqrt{2}$$

$$-6\sqrt{2} = w\sqrt{2}$$

$$w = -6$$

$$w = 6$$

Find X. Find y.

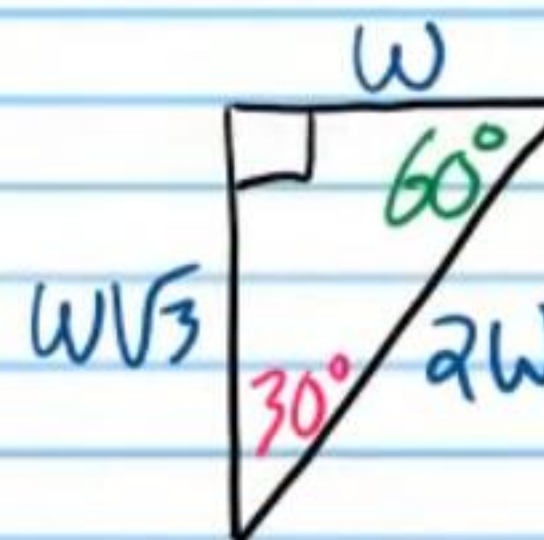
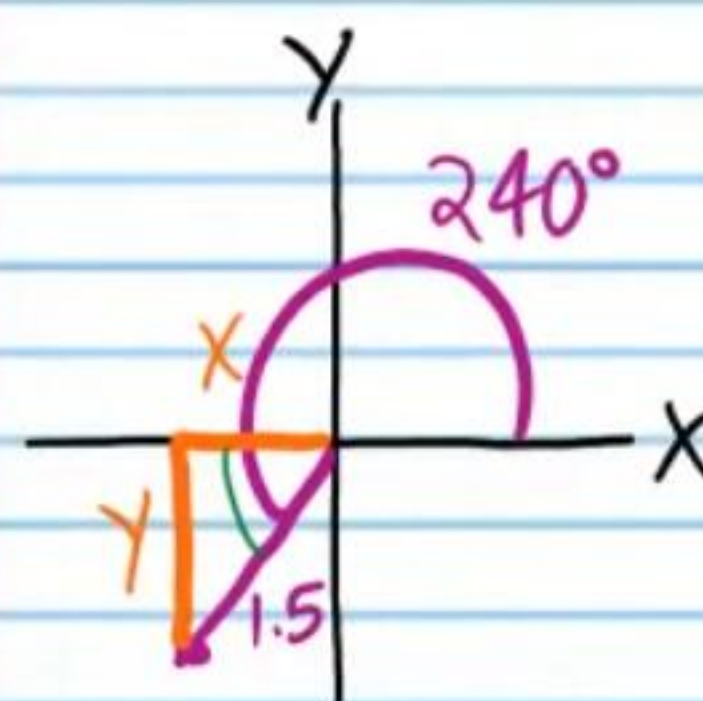
$$x = -w \quad y = w$$

$$x = -6 \quad y = 6$$

$$(-6, 6)$$

⑤④ $(1.5, 240^\circ)$

Graph



$$r = 2w$$

$$1.5 = 2w$$

$$w = \frac{3}{4}$$

Find X. Find y.

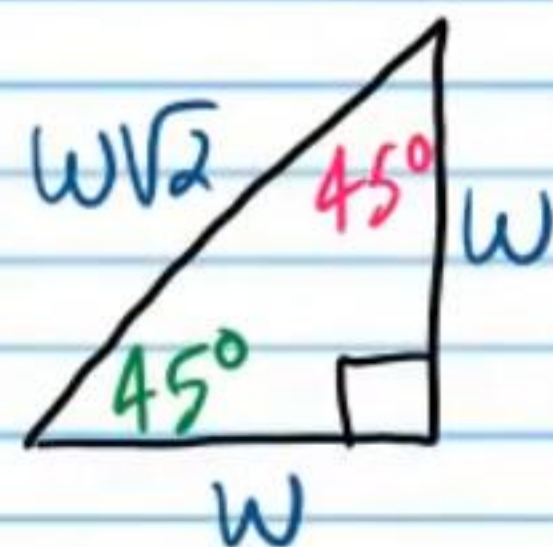
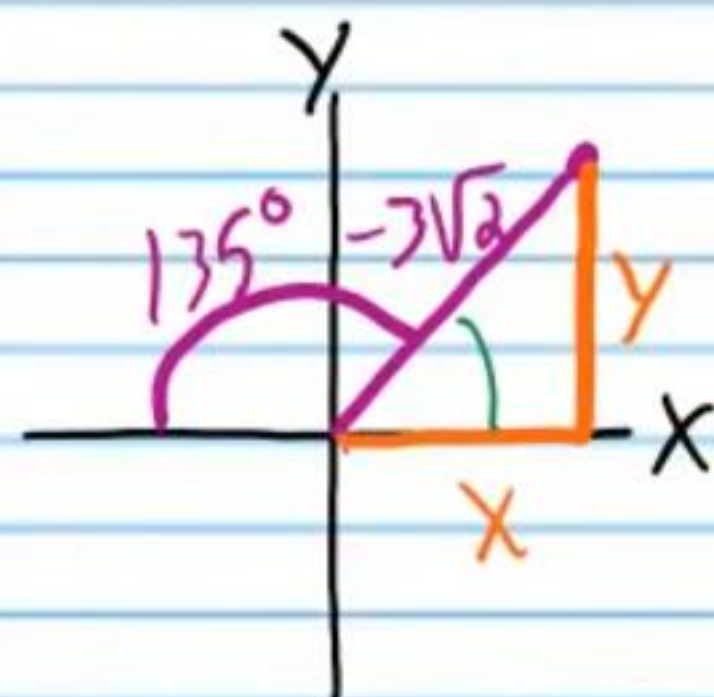
$$x = -w \quad y = -w\sqrt{3}$$

$$x = -\frac{3}{4} \quad y = -\frac{3}{4}\sqrt{3}$$

$$\left(-\frac{3}{4}, -\frac{3}{4}\sqrt{3}\right)$$

⑤④ $(-3\sqrt{2}, -135^\circ)$

Graph



$$r = W\sqrt{2}$$

$$-3\sqrt{2} = W\sqrt{2}$$

$$W = -3$$

$$W = 3$$

$$\boxed{(3, 3)}$$

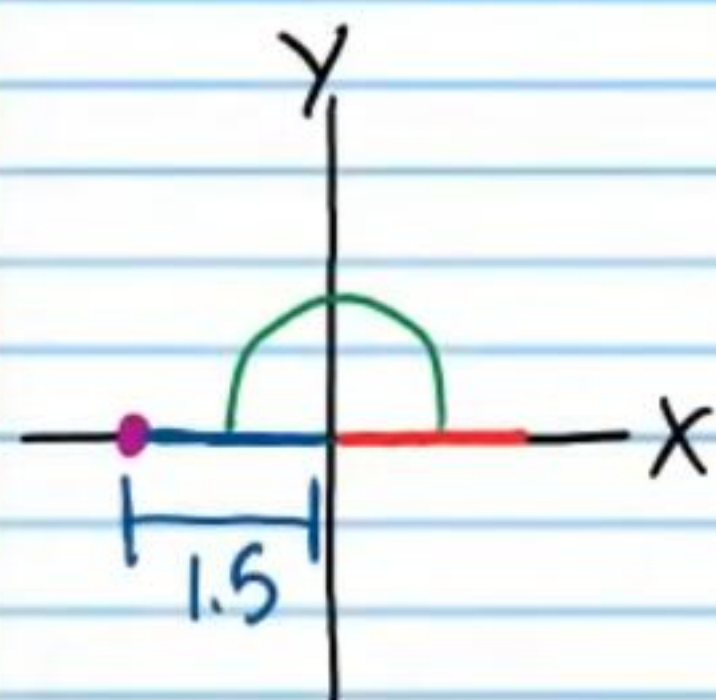
Find X. Find y.

$$x = W \quad y = W$$

$$x = 3 \quad y = 3$$

• ⑤⑤ $(1.5, 180^\circ)$

Graph



Find X. Find y.

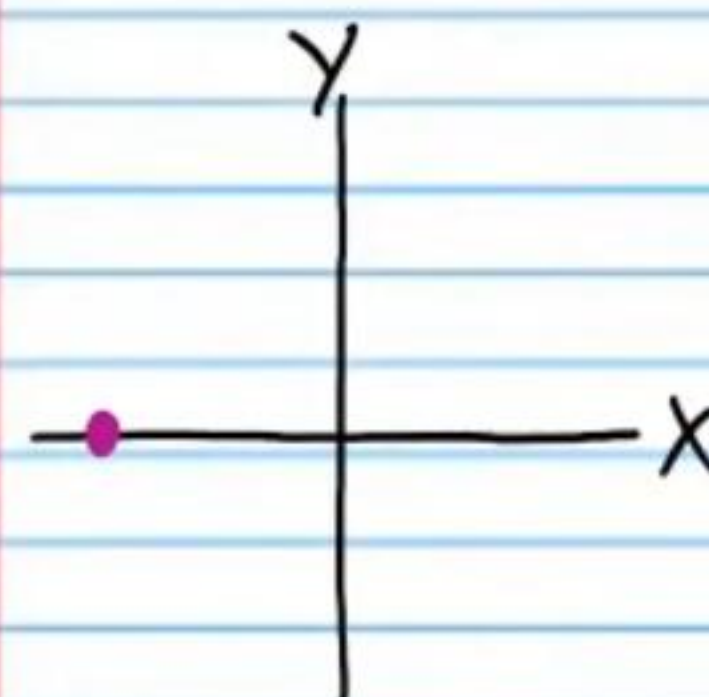
$$x = -1.5$$

$$y = 0$$

$$\boxed{(-1.5, 0)}$$

⑤⑥ $(-2, 0^\circ)$

Graph



Find X. Find y.

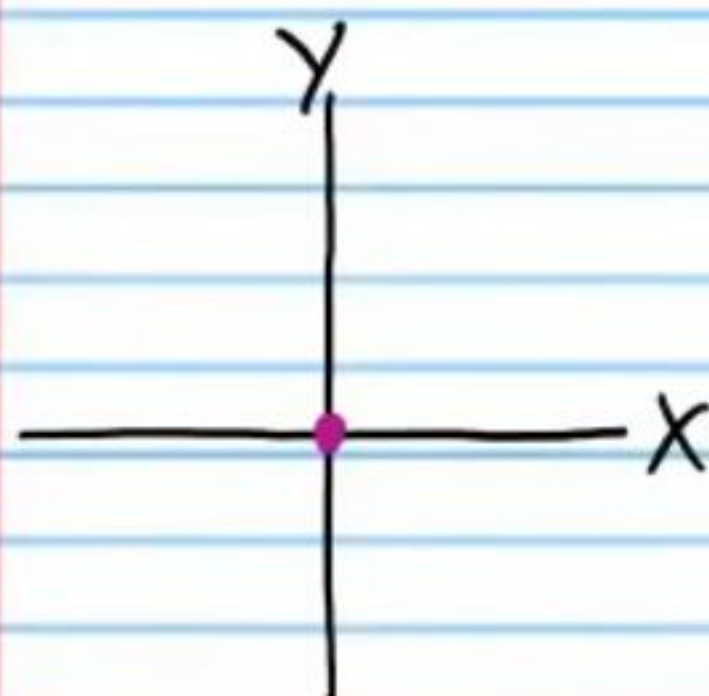
$$x = -2$$

$$y = 0$$

$$\boxed{(-2, 0)}$$

⑤⑦ $(0, 24.67^\circ)$

Graph



Find X. Find y.

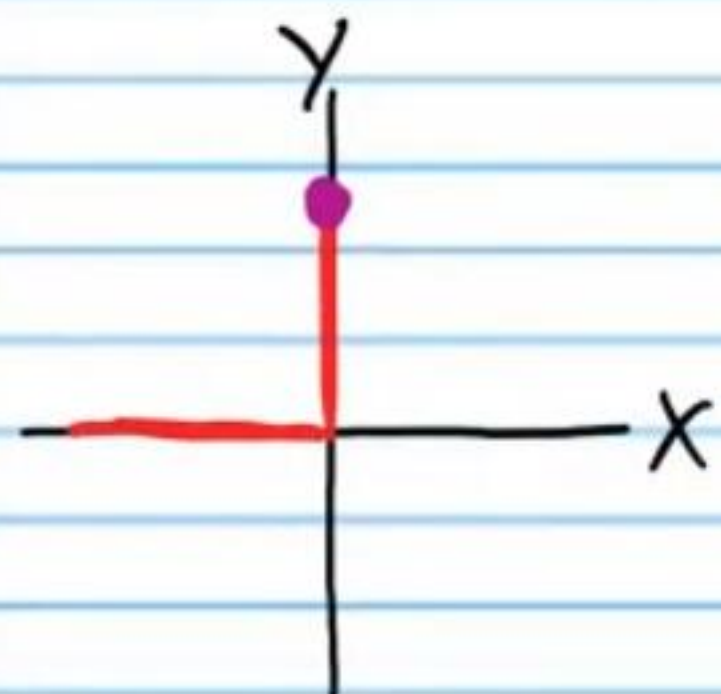
$$x = 0$$

$$y = 0$$

$$\boxed{(0, 0)}$$

• (58) $(-3, -90^\circ)$

Graph



Find X. Find y.

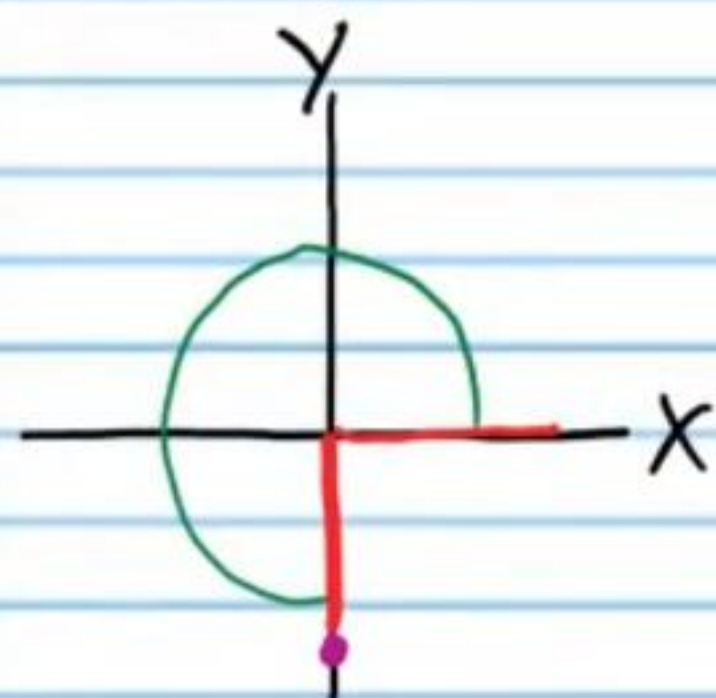
$$x=0$$

$$y=3$$

$$(0, 3)$$

(59) $(1, 270^\circ)$

Graph



Find X. Find y.

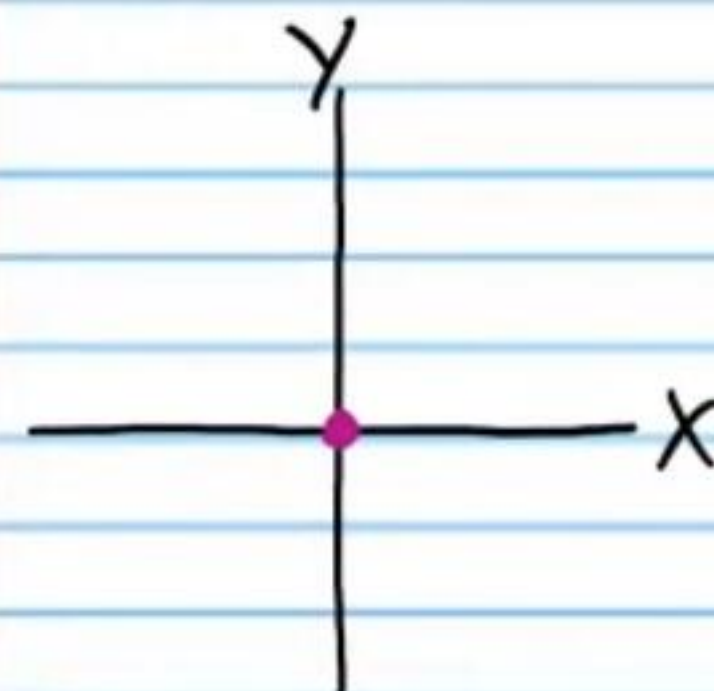
$$x=0$$

$$y=-1$$

$$(0, -1)$$

(60) $(0, 172^\circ)$

Graph



Find X. Find y.

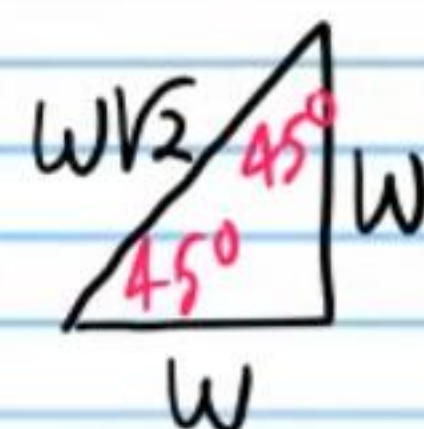
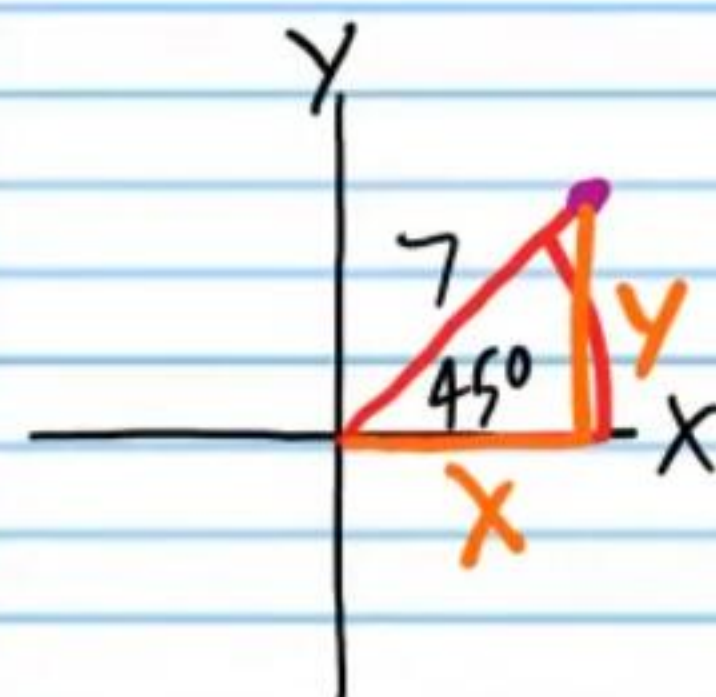
$$x=0$$

$$y=0$$

$$(0, 0)$$

• (61) $(7, 45^\circ)$

Graph



Find X. Find y.

$$x=W$$

$$y=W$$

$$x = \frac{7\sqrt{2}}{2}$$

$$y = \frac{7\sqrt{2}}{2}$$

$$r = W\sqrt{2}$$

$$7 = W\sqrt{2}$$

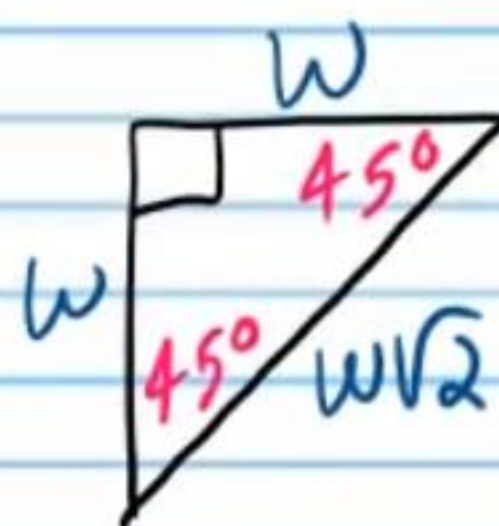
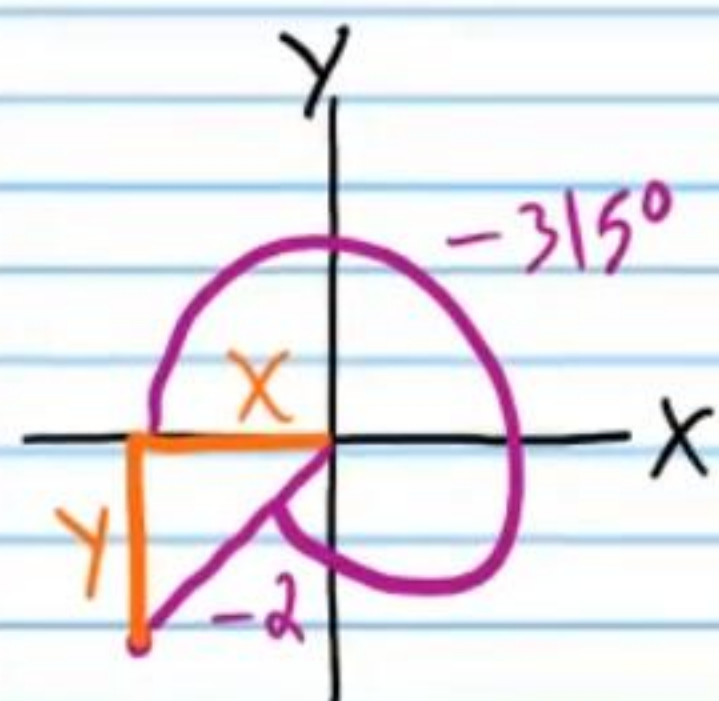
$$\frac{7}{\sqrt{2}} = W$$

$$\frac{7\sqrt{2}}{2} = W$$

$$\left(\frac{7\sqrt{2}}{2}, \frac{7\sqrt{2}}{2} \right)$$

• 6d) $(-2, -315^\circ)$

Graph



Find x . Find y .

$$x = -w \quad y = -w$$

$$x = -\sqrt{2} \quad y = -\sqrt{2}$$

$$r = w\sqrt{2}$$

$$-2 = w\sqrt{2}$$

$$\frac{-2}{\sqrt{2}} = w$$

$$\frac{-2\sqrt{2}}{2} = w$$

$$-\sqrt{2} = w$$

$$\sqrt{2} = w$$

$$(-\sqrt{2}, -\sqrt{2})$$